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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Atoms and Nuclei

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 180 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Two radioactive substances A and B of mass numbers 200 and 212 respectively, shows spontaneous α-decay with same Q value of 1 MeV . The ratio of energies of α-rays produced by A and B is ____ .

(A) $\frac{2548}{2650}$ (B) $\frac{2706}{2646}$ (C) $\frac{2597}{2600}$ (D) $\frac{2862}{2499}$

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

The energy released if hydrogen atoms are combined to form ${}_{2}^{4}\mathrm{He}$ is ____ MeV .

(Take binding energies per nucleon of ${}_{1}^{2}\mathrm{H}$ and ${}_{2}^{4}\mathrm{He}$ as 1.1 MeV and 7.2 MeV , respectively)

(A) 6.1 (B) 24.4 (C) 26.6 (D) 5

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β. The possible values of α and β are:-

(A) 27 and 20 (B) 3 and 16 (C) 5 and 36 (D) 20 and 27

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Assuming the experimental mass of ${}_{6}^{12}C$ as $12u$, the mass defect of ${}_{6}^{12}C$ atom is ____ $\mathrm{MeV/c^2}$.

(Mass of proton $= 1.00727\mathrm{u}$. mass of neutron $= 1.00866\mathrm{u}, 1\mathrm{u} = 931.5\mathrm{MeV/c^2}$ and c is the speed of the light in vacuum).

(A) 127.5 (B) 89.03 (C) 272.0 (D) 92.0

Open solution

Q5. JEE Main 2026 (Online) 5th April Morning Shift

In the hydrogen atom, the electron makes a transition from the higher orbit (i) to a lower orbit (f). The ratio of the radius of the orbits in given by $r_i : r_f = 16 : 4$. The wavelength of photon emitted due to this transition is ____ nm.

(Given Rydberg constant $= 1.0973 \times 10^7/\mathrm{m}$)

(A) 121 (B) 242 (C) 486 (D) 974

Open solution

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Q6. JEE Main 2026 (Online) 4th April Evening Shift

In Rutherford's alpha-particle scattering experiment, only a few alpha particles rebound back because

A. The size of gold nucleus is very small as compared to the size of gold atom.

B. Alpha particle and gold nucleus have equal charge.

C. The impact parameter is minimum for a few alpha particles.

D. A few alpha particles have very high kinetic energy.

E. Only a few alpha particles undergo head-on collision with the nuclei.

Choose the correct answer from the options given below :

(A) A, B Only (B) B, E Only (C) C, D Only (D) A, C, E Only

Open solution

Q7. JEE Main 2026 (Online) 4th April Morning Shift

Two nuclei of mass number 3 combine with another nucleus of mass number 4 to yield a nucleus of mass number 10. If the binding energy per nucleon for the mass numbers 3,4 and 10 are $5.6\text{MeV}, 7.4\text{MeV}$ and 6.1 MeV , respectively, then in the process, $\Delta\text{Mc}^2 =$ ____ MeV .

(A) 6.9 (B) 7.9 (C) 2.2 (D) 4.3

Open solution

Q8. JEE Main 2026 (Online) 2nd April Evening Shift

The binding energy per nucleon of $^{209}_{83}Bi$ is _____ MeV.

[Take $m(^{209}_{83}Bi) = 208.980388$ u, $m_p = 1.007825$ u, $m_n = 1.008665$ u, 1 u $= 931$ MeV$/c^2$]

(A) 7.48 (B) 7.84 (C) 8.79 (D) 6.94

Open solution

Q9. JEE Main 2026 (Online) 2nd April Morning Shift

Angular momentum of an electron in a hydrogen atom is $\frac{3h}{\pi}$, then the energy of the electron is ____ eV.

(A) -1.51 (B) -0.85 (C) -0.38 (D) -0.28

Open solution

Q10. JEE Main 2026 (Online) 28th January Evening Shift

A nucleus has mass number α and radius R_α. Another nucleus has mass number β and radius R_β.

If $\beta = 8\alpha$ then R_α/R_β is :

(A) 1 (B) 2 (C) 8 (D) 0.5

Open solution

Q11. JEE Main 2026 (Online) 28th January Morning Shift

An atom $^{8}_{3}X$ is bombarded by shower of fundamental particles and in 10 s this atom absorbed 10 electrons, 10 protons and 9 neutrons. The percentage growth in the surface area of the nucleons is recorded by :

(A) 150% (B) 900% (C) 125% (D) 225%

Open solution

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Q12. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

The binding energy for the following nuclear reactions are expressed in MeV .

$_2He^3 + {}_0n^1 \rightarrow {}_2He^4 + 20MeV$

$_2He^4 + {}_0n^1 \rightarrow {}_2He^5 - 0.9MeV$

If X_3, X_4, X_5 denote the stability of $_2He^3$, $_2He^4$ and $_2He^5$, respectively, then the correct order is :

(A) $X_4 > X_5 > X_3$ (B) $X_4 < X_5 < X_3$ (C) $X_4 > X_5 < X_3$ (D) $X_4 = X_5 = X_3$

Q13. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Two electrons are moving in orbits of two hydrogen like atoms with speeds 3×10^5 m/s and 2.5×10^5 m/s respectively. If the radii of these orbits are nearly same then the possible order of energy states are ____ respectively.

(A) 8 and 10 (B) 10 and 12 (C) 9 and 8 (D) 6 and 5

Q14. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Given below are two statements :

Statement I : For all elements, greater the mass of the nucleus, greater is the binding energy per nucleon.

Statement II : For all elements, nuclei with less binding energy per nucleon transforms to nuclei with greater binding energy per nucleon.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is true but Statement II is false

Q15. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

Which of the following pair of nuclei are isobars of the element?

(A) ^3_1H and ^3_2He (B) $^{198}_{80}Hg$ and $^{197}_{79}Au$

(C) ^2_1H and ^3_1H (D) $^{236}_{92}U$ and $^{238}_{92}U$

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Q16. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

In hydrogen atom spectrum, ($R \rightarrow$ Rydberg's constant)

A. the maximum wavelength of the radiation of Lyman series is $\frac{4}{3R}$

B. the Balmer series lies in the visible region of the spectrum

C. the minimum wavelength of the radiation of Paschen series is $\frac{9}{R}$

D. the minimum wavelength of Lyman series is $\frac{5}{4R}$

Choose the correct answer from the options given below :

(A) A, B Only (B) B, D Only (C) A, B and D Only (D) A, B and C Only

Q17. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

The smallest wavelength of Lyman series is 91 nm . The difference between the largest wavelengths of Paschen and Balmer series is nearly ____ nm.

(A) 1784 (B) 1875 (C) 1217 (D) 1550

Q18. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

The minimum frequency of photon required to break a particle of mass 15.348 amu into 4α particles is ____ kHz .

[mass of He nucleus = $4.002\text{amu}, 1\text{amu} = 1.66 \times 10^{-27}\ \text{kg},\ \text{h} = 6.6 \times 10^{-34}\ \text{J.s}$ and $\text{c} = 3 \times 10^{8}\ \text{m/s}$]

(A) 14.94×10^{20} (B) 9×10^{19} (C) 9×10^{20} (D) 14.94×10^{19}

Q19. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

$7.9\text{MeV}\ \alpha$-particle scatters from a target material of atomic number 79 . From the given data the estimated diameter of nuclei of the target material is (approximately) ____ m.

$\left[\frac{1}{4\pi\epsilon_o} = 9 \times 10^{9}\text{Nm}^2/\text{C}^2 \text{ and electron charge } = 1.6 \times 10^{-19}\text{C}\right]$

(A) 2.88×10^{-14} (B) 5.76×10^{-14} (C) 1.44×10^{-13} (D) 1.69×10^{-12}

Q20. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

The energy of an electron in an orbit of the Bohr's atom is $-0.04E_0$ eV where E_0 is the ground state energy. If L is the angular momentum of the electron in this orbit and h is the Planck's constant, then

$\frac{2\pi L}{h}$ is _____ :

(A) 6 (B) 2 (C) 5 (D) 4

Q21. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

If an alpha particle with energy 7.7 MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is ____ m. (Atomic number of gold $= 79$ and $\frac{1}{4\pi\epsilon_o} = 9 \times 10^{9}$ in SI units)

(A) 2.95×10^{-16} (B) 3.85×10^{-14} (C) 2.95×10^{-14} (D) 3.85×10^{-16}

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Q22. JEE Main 2025 (Online) 8th April Evening Shift

For a nucleus of mass number A and radius R, the mass density of nucleus can be represented as

(A) $A^{\frac{2}{3}}$ (B) Independent of A (C) A^3 (D) $A^{\frac{1}{3}}$

Open solution

Q23. JEE Main 2025 (Online) 7th April Evening Shift

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The density of the copper $\left({}^{64}_{29}\mathrm{Cu}\right)$ nucleus is greater than that of the carbon $\left({}^{12}_{6}\mathrm{C}\right)$ nucleus.

Reason (R) : The nucleus of mass number A has a radius proportional to $A^{1/3}$.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(A) (A) is correct but (R) is not correct

(B) Both (A) and (R) are correct but (R) is **not** the correct explanation of (A)

(C) (A) is not correct but (R) is correct

(D) Both (A) and (R) are correct and (R) **is** the correct explanation of (A)

Open solution

Q24. JEE Main 2025 (Online) 7th April Morning Shift

In a hydrogen like ion, the energy difference between the 2nd excitation energy state and ground is 108.8 eV . The atomic number of the ion is:

(A) 1 (B) 4 (C) 3 (D) 2

Open solution

Q25. JEE Main 2025 (Online) 7th April Morning Shift

For a hydrogen atom, the ratio of the largest wavelength of Lyman series to that of the Balmer series is

(A) $5:27$ (B) $27:5$ (C) $3:4$ (D) $5:36$

Open solution

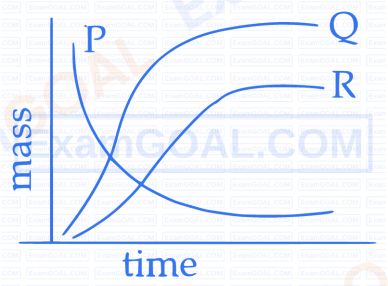
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Q26. JEE Main 2025 (Online) 4th April Evening Shift

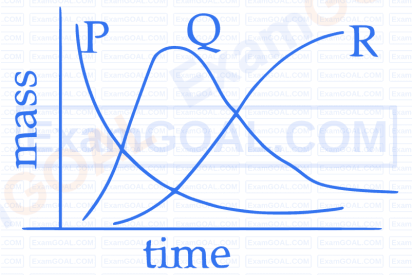
Open solution

A radioactive material P first decays into Q and then Q decays to non-radioactive material R. Which of the following figure represents time dependent mass of P, Q and R ?

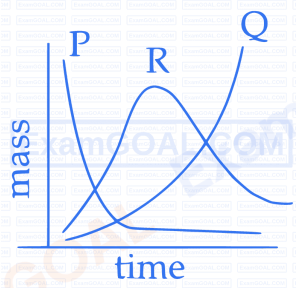
(A)



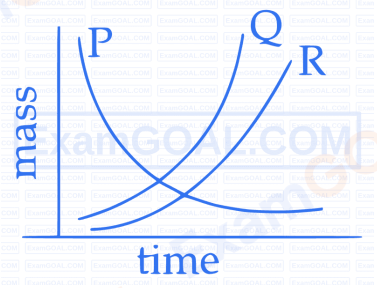
(B)



(C)



(D)



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Q27. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Given below are two statements :

Statement (I) : The dimensions of Planck's constant and angular momentum are same.

Statement (II) : In Bohr's model electron revolve around the nucleus only in those orbits for which angular momentum is integral multiple of Planck's constant.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but Statement II is correct

Q28. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Considering the Bohr model of hydrogen like atoms, the ratio of the radius of 5th orbit of the electron in Li^{2+} and $He^{+}$is

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{4}{9}$ (D) $\frac{9}{4}$

Q29. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Given below are two statements: one is labelled as $\mathbf{Assertion A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : The Bohr model is applicable to hydrogen and hydrogen-like atoms only.

Reason $\mathbf{R}$: The formulation of Bohr model does not include repulsive force between electrons.

In the light of the above statements, choose the correct answer from the options given below

(A) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

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Q30. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Match the LIST-I with LIST-II

List - I		List - II	
A.	${}_0^1n + {}_{92}^{235}U \rightarrow {}_{54}^{140}Xe + {}_{38}^{94}Sr + 2{}_0^1n$	I.	Chemical reaction
B.	$2H_2 + O_2 \rightarrow 2H_2O$	II.	Fusion with +ve Q value
C.	${}_1^2H + {}_1^2H \rightarrow {}_2^3He + {}_0^1n$	III.	Fission
D.	${}_1^1H + {}_1^3H \rightarrow {}_1^2H + {}_1^2H$	IV.	Fusion with -ve Q value

Choose the correct answer from the options given below:

(A) A-II, B-I, C-III, D-IV

(B) A-III, B-I, C-II, D-IV

(C) A-III, B-I, C-IV, D-II

(D) A-II, B-I, C-IV, D-III

Q31. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Energy released when two deuterons $\left({}_1H^2\right)$ fuse to form a helium nucleus $\left({}_2He^4\right)$ is : (Given : Binding energy per nucleon of ${}_1H^2 = 1.1MeV$ and binding energy per nucleon of ${}_2He^4 = 7.0MeV$)

(A) 26.8 MeV

(B) 8.1 MeV

(C) 23.6 MeV

(D) 5.9 MeV

Q32. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Assuming the validity of Bohr's atomic model for hydrogen like ions the radius of Li^{++} ion in its ground state is given by $\frac{1}{X}a_0$, where $X =$ _____ (Where a_0 is the first Bohr's radius.)

(A) 2

(B) 9

(C) 1

(D) 3

Q33. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

Considering Bohr's atomic model for hydrogen atom :

(A) the energy of H atom in ground state is same as energy of He^+ ion in its first excited state.

(B) the energy of H atom in ground state is same as that for Li^{++} ion in its second excited state.

(C) the energy of H atom in its ground state is same as that of He^+ ion for its ground state.

(D) the energy of He^+ ion in its first excited state is same as that for Li^{++} ion in its ground state.

Choose the correct answer from the options given below :

(A) (A), (B) only

(B) (A), (D) only

(C) (A), (C) only

(D) (B), (D) only

Q34. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

The number of spectral lines emitted by atomic hydrogen that is in the 4th energy level, is

(A) 3

(B) 6

(C) 1

(D) 0

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Q35. JEE Main 2025 (Online) 28th January Evening Shift

The frequency of revolution of the electron in Bohr's orbit varies with *n*, the principal quantum number as:

(A) $\frac{1}{n^4}$ (B) $\frac{1}{n^2}$ (C) $\frac{1}{n^3}$ (D) $\frac{1}{n}$

Open solution

Q36. JEE Main 2025 (Online) 28th January Morning Shift

Choose the correct nuclear process from the below options [p : proton, n : neutron, e^-: electron, e^+: positron, v : neutrino, $\bar{v}$: antineutrino]

(A) $n \rightarrow p + e^+ + v$ (B) $n \rightarrow p + e^- + \bar{v}$

(C) $n \rightarrow p + e^+ + \bar{v}$ (D) $n \rightarrow p + e^- + v$

Open solution

Q37. JEE Main 2025 (Online) 24th January Evening Shift

The energy E and momentum p of a moving body of mass m are related by some equation. Given that c represents the speed of light, identify the correct equation

(A) $E^2 = pc^2 + m^2c^4$ (B) $E^2 = pc^2 + m^2c^2$

(C) $E^2 = p^2c^2 + m^2c^4$ (D) $E^2 = p^2c^2 + m^2c^2$

Open solution

Q38. JEE Main 2025 (Online) 24th January Morning Shift

During the transition of electron from state A to state C of a Bohr atom, the wavelength of emitted radiation is $2000\overset{o}{A}$ and it becomes $6000\overset{o}{A}$ when the electron jumps from state B to state C. Then the wavelength of the radiation emitted during the transition of electrons from state A to state B is

(A) $4000\overset{o}{A}$ (B) $6000\overset{o}{A}$ (C) $2000\overset{o}{A}$ (D) $3000\overset{o}{A}$

Open solution

Q39. JEE Main 2025 (Online) 23rd January Evening Shift

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The binding energy per nucleon is found to be practically independent of the atomic number A , for nuclei with mass numbers between 30 and 170 .

Reason (R) : Nuclear force is long range. In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

Open solution

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Q40. JEE Main 2025 (Online) 23rd January Morning Shift

A radioactive nucleus n_2 has 3 times the decay constant as compared to the decay constant of another radioactive nucleus n_1. If initial number of both nuclei are the same, what is the ratio of number of nuclei of n_2 to the number of nuclei of n_1, after one half-life of n_1 ?

(A) 1/4 (B) 1/8 (C) 4 (D) 8

Open solution

Q41. JEE Main 2024 (Online) 9th April Evening Shift

A nucleus at rest disintegrates into two smaller nuclei with their masses in the ratio of $2:1$. After disintegration they will move :

(A) in opposite directions with speed in the ratio of $1:2$ respectively.

(B) in the same direction with same speed.

(C) in opposite directions with speed in the ratio of $2:1$ respectively.

(D) in opposite directions with the same speed.

Open solution

Q42. JEE Main 2024 (Online) 9th April Evening Shift

The energy released in the fusion of $2\ \text{kg}$ of hydrogen deep in the sun is E_H and the energy released in the fission of $2\ \text{kg}$ of ^{235}U is E_U. The ratio $\frac{E_H}{E_U}$ is approximately: (Consider the fusion reaction as $4_1^1H + 2e^- \rightarrow {}_2^4\text{He} + 2v + 6\gamma + 26.7\ \text{MeV}$, energy released in the fission reaction of ^{235}U is $200\ \text{MeV}$ per fission nucleus and $\text{N}_\text{A} = 6.023 \times 10^{23}$)

(A) 7.62 (B) 25.6 (C) 9.13 (D) 15.04

Open solution

Q43. JEE Main 2024 (Online) 9th April Evening Shift

A hydrogen atom in ground state is given an energy of $10.2\ \text{eV}$. How many spectral lines will be emitted due to transition of electrons?

(A) 3 (B) 6 (C) 10 (D) 1

Open solution

Q44. JEE Main 2024 (Online) 9th April Morning Shift

The energy equivalent of $1\ \text{g}$ of substance is :

(A) $5.6 \times 10^{26}\ \text{MeV}$ (B) $5.6 \times 10^{12}\ \text{MeV}$

(C) $5.6\ \text{eV}$ (D) $11.2 \times 10^{24}\ \text{MeV}$

Open solution

Q45. JEE Main 2024 (Online) 8th April Evening Shift

If M_0 is the mass of isotope ${}_5^{12}B$, M_p and M_n are the masses of proton and neutron, then nuclear binding energy of isotope is:

(A) $(5M_p + 7M_n - M_0)C^2$ (B) $(M_0 - 5M_p - 7M_n)C^2$

(C) $(M_0 - 5M_p)C^2$ (D) $(M_0 - 12M_n)C^2$

Open solution

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Q46. JEE Main 2024 (Online) 8th April Evening Shift

In a hypothetical fission reaction

$_{92}X^{236} \rightarrow {}_{56}\mathrm{Y}^{141} + {}_{36}Z^{92} + 3R$

The identity of emitted particles (R) is :

(A) Proton (B) Neutron (C) Electron (D) γ-radiations

Open solution

Q47. JEE Main 2024 (Online) 8th April Morning Shift

Binding energy of a certain nucleus is 18×10^8 J. How much is the difference between total mass of all the nucleons and nuclear mass of the given nucleus:

(A) 20 μg (B) 2 μg (C) 10 μg (D) 0.2 μg

Open solution

Q48. JEE Main 2024 (Online) 6th April Evening Shift

The longest wavelength associated with Paschen series is : (Given $\mathrm{R_H} = 1.097 \times 10^7 \mathrm{SI}$ unit)

(A) $2.973 \times 10^{-6}\ \mathrm{m}$ (B) $1.876 \times 10^{-6}\ \mathrm{m}$ (C) $1.094 \times 10^{-6}\ \mathrm{m}$ (D) $3.646 \times 10^{-6}\ \mathrm{m}$

Open solution

Q49. JEE Main 2024 (Online) 6th April Morning Shift

The ratio of the shortest wavelength of Balmer series to the shortest wavelength of Lyman series for hydrogen atom is :

(A) $1:2$ (B) $1:4$ (C) $2:1$ (D) $4:1$

Open solution

Q50. JEE Main 2024 (Online) 5th April Evening Shift

The angular momentum of an electron in a hydrogen atom is proportional to : (Where r is the radius of orbit of electron)

(A) $\frac{1}{\mathrm{r}}$ (B) $\frac{1}{\sqrt{\mathrm{r}}}$ (C) $\sqrt{\mathrm{r}}$ (D) r

Open solution

Q51. JEE Main 2024 (Online) 5th April Morning Shift

An electron rotates in a circle around a nucleus having positive charge Ze. Correct relation between total energy (E) of electron to its potential energy (U) is :

(A) $2\mathrm{E} = 3\mathrm{U}$ (B) $\mathrm{E} = \mathrm{U}$ (C) $2\mathrm{E} = \mathrm{U}$ (D) $\mathrm{E} = 2\mathrm{U}$

Open solution

Q52. JEE Main 2024 (Online) 4th April Evening Shift

According to Bohr's theory, the moment of momentum of an electron revolving in 4th orbit of hydrogen atom is:

(A) $2\frac{h}{\pi}$ (B) $\frac{h}{2\pi}$ (C) $\frac{h}{\pi}$ (D) $8\frac{h}{\pi}$

Open solution

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Q53. JEE Main 2024 (Online) 4th April Morning Shift

Which of the following nuclear fragments corresponding to nuclear fission between neutron $\left({}_0^1\mathrm{n}\right)$ and uranium isotope $\left({}_{92}^{235}\mathrm{U}\right)$ is correct :

Open solution

(A) ${}_{56}^{140}\mathrm{Xe} + {}_{38}^{94}\mathrm{Sr} + 3{}_0^1\mathrm{n}$

(B) ${}_{51}^{153}\mathrm{Sb} + {}_{41}^{99}\mathrm{Nb} + 3{}_0^1\mathrm{n}$

(C) ${}_{56}^{144}\mathrm{Ba} + {}_{36}^{89}\mathrm{Kr} + 4{}_0^1\mathrm{n}$

(D) ${}_{56}^{144}\mathrm{Ba} + {}_{36}^{89}\mathrm{Kr} + 3{}_0^1\mathrm{n}$

Q54. JEE Main 2024 (Online) 1st February Evening Shift

From the statements given below :

Open solution

(A) The angular momentum of an electron in n^{th} orbit is an integral multiple of $\hbar$.

(B) Nuclear forces do not obey inverse square law.

(C) Nuclear forces are spin dependent.

(D) Nuclear forces are central and charge independent.

(E) Stability of nucleus is inversely proportional to the value of packing fraction.

Choose the **correct** answer from the options given below :

(A) (B), (C), (D), (E) only

(B) (A), (C), (D), (E) only

(C) (A), (B), (C), (E) only

(D) (A), (B), (C), (D) only

Q55. JEE Main 2024 (Online) 1st February Morning Shift

The minimum energy required by a hydrogen atom in ground state to emit radiation in Balmer series is nearly :

Open solution

(A) $13.6\mathrm{eV}$

(B) $1.5\mathrm{eV}$

(C) $12.1\mathrm{eV}$

(D) $1.9\mathrm{eV}$

Q56. JEE Main 2024 (Online) 31st January Evening Shift

The mass number of nucleus having radius equal to half of the radius of nucleus with mass number 192 is :

Open solution

(A) 32

(B) 24

(C) 20

(D) 40

Q57. JEE Main 2024 (Online) 31st January Morning Shift

If the wavelength of the first member of Lyman series of hydrogen is λ. The wavelength of the second member will be

Open solution

(A) $\frac{27}{5}\lambda$

(B) $\frac{5}{27}\lambda$

(C) $\frac{27}{32}\lambda$

(D) $\frac{32}{27}\lambda$

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Q58. JEE Main 2024 (Online) 30th January Evening Shift

An electron revolving in n^{th} Bohr orbit has magnetic moment μ_n. If $\mu_n \propto n^x$, the value of x is

(A) 2 (B) 0 (C) 3 (D) 1

Open solution

Q59. JEE Main 2024 (Online) 30th January Evening Shift

In a nuclear fission reaction of an isotope of mass M, three similar daughter nuclei of same mass are formed. The speed of a daughter nucleus in terms of mass defect ΔM will be :

(A) $c\sqrt{\frac{3\Delta M}{M}}$ (B) $\frac{\Delta Mc^2}{3}$ (C) $c\sqrt{\frac{2\Delta M}{M}}$ (D) $\sqrt{\frac{2c\Delta M}{M}}$

Open solution

Q60. JEE Main 2024 (Online) 30th January Morning Shift

The ratio of the magnitude of the kinetic energy to the potential energy of an electron in the 5th excited state of a hydrogen atom is :

(A) 4 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Open solution

Q61. JEE Main 2024 (Online) 29th January Evening Shift

Given below are two statements:

Statement I : Most of the mass of the atom and all its positive charge are concentrated in a tiny nucleus and the electrons revolve around it, is Rutherford's model.

Statement II : An atom is a spherical cloud of positive charges with electrons embedded in it, is a special case of Rutherford's model.

In the light of the above statements, choose the most appropriate from the options given below

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both statement I and statement II are false

Open solution

Q62. JEE Main 2024 (Online) 29th January Morning Shift

The explosive in a Hydrogen bomb is a mixture of ${}_1H^2$, ${}_1H^3$ and ${}_3Li^6$ in some condensed form. The chain reaction is given by

${}_3Li^6 + {}_0n^1 \rightarrow {}_2He^4 + {}_1H^3$

${}_1H^2 + {}_1H^3 \rightarrow {}_2He^4 + {}_0n^1$

During the explosion the energy released is approximately

[Given ; $M(Li) = 6.01690\ amu, M\left({}_1H^2\right) = 2.01471\ amu, M\left({}_2He^4\right) = 4.00388\ amu$, and $1\ amu = 931.5\ MeV$]

(A) 22.22 MeV (B) 28.12 MeV (C) 16.48 MeV (D) 12.64 MeV

Open solution

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Q63. JEE Main 2024 (Online) 27th January Evening Shift

The atomic mass of ${}_6C^{12}$ is $12.000000\ u$ and that of ${}_6C^{13}$ is $13.003354\ u$. The required energy to remove a neutron from ${}_6C^{13}$, if mass of neutron is $1.008665\ u$, will be :

(A) 62.5 MeV (B) 6.25 MeV (C) 4.95 MeV (D) 49.5 MeV

Open solution

Q64. JEE Main 2024 (Online) 27th January Morning Shift

The radius of third stationary orbit of electron for Bohr's atom is R. The radius of fourth stationary orbit will be:

(A) $\frac{4}{3}R$ (B) $\frac{16}{9}R$ (C) $\frac{3}{4}R$ (D) $\frac{9}{16}R$

Open solution

Q65. JEE Main 2023 (Online) 13th April Morning Shift

$${}^{238}_{92}A \rightarrow {}^{234}_{90}B + {}^{4}_{2}D + Q$$

In the given nuclear reaction, the approximate amount of energy released will be:

[Given, mass of ${}^{238}_{92}A = 238.05079 \times 931.5\ MeV/c^2$,

mass of ${}^{234}_{90}B = 234.04363 \times 931.5\ MeV/c^2$,

mass of ${}^{4}_{2}D = 4.00260 \times 931.5\ MeV/c^2$]

(A) 2.12 MeV (B) 4.25 MeV (C) 3.82 MeV (D) 5.9 MeV

Open solution

Q66. JEE Main 2023 (Online) 12th April Morning Shift

A $12.5\ eV$ electron beam is used to bombard gaseous hydrogen at room temperature. The number of spectral lines emitted will be:

(A) 2 (B) 4 (C) 3 (D) 1

Open solution

Q67. JEE Main 2023 (Online) 11th April Evening Shift

The energy of He^+ ion in its first excited state is, (The ground state energy for the Hydrogen atom is $-13.6\ eV$) :

(A) $-13.6\ eV$ (B) $-27.2\ eV$ (C) $-3.4\ eV$ (D) $-54.4\ eV$

Open solution

Q68. JEE Main 2023 (Online) 10th April Morning Shift

The angular momentum for the electron in Bohr's orbit is L. If the electron is assumed to revolve in second orbit of hydrogen atom, then the change in angular momentum will be

(A) L (B) $\frac{L}{2}$ (C) zero (D) 2 L

Open solution

Q69. JEE Main 2023 (Online) 8th April Evening Shift

The waves emitted when a metal target is bombarded with high energy electrons are

(A) Infrared rays (B) Radio Waves (C) Microwaves (D) X-rays

Open solution

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Q70. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

For a nucleus ${}^{A}_{Z}X$ having mass number A and atomic number Z

A. The surface energy per nucleon $(b_s) = -a_1 A^{2/3}$.

B. The Coulomb contribution to the binding energy $b_c = -a_2 \frac{Z(Z-1)}{A^{4/3}}$

C. The volume energy $b_v = a_3 A$

D. Decrease in the binding energy is proportional to surface area.

E. While estimating the surface energy, it is assumed that each nucleon interacts with 12 nucleons. (a_1, a_2 and a_3 are constants)

Choose the most appropriate answer from the options given below:

(A) C, D only (B) B, C, E only (C) B, C only (D) A, B, C, D only

Q71. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

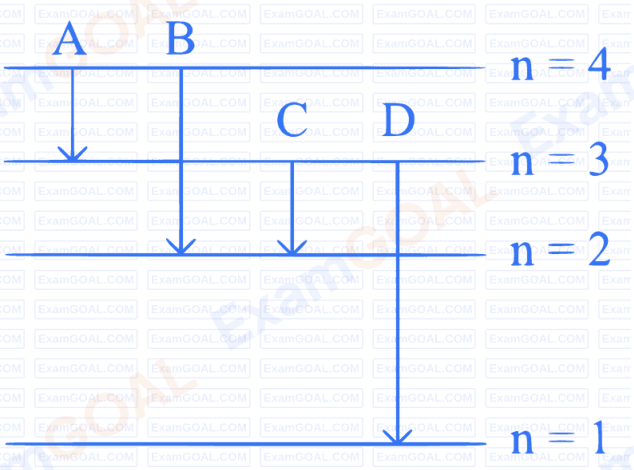
A small particle of mass m moves in such a way that its potential energy $U = \frac{1}{2} m \omega^2 r^2$ where ω is constant and r is the distance of the particle from origin. Assuming Bohr's quantization of momentum and circular orbit, the radius of nth orbit will be proportional to,

(A) $\sqrt{n}$ (B) n^2 (C) $\frac{1}{n}$ (D) n

Q72. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

The energy levels of an hydrogen atom are shown below. The transition corresponding to emission of shortest wavelength is :



(A) A (B) C (C) B (D) D

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Q73. JEE Main 2023 (Online) 1st February Evening Shift

An electron of a hydrogen like atom, having $Z = 4$, jumps from 4th energy state to 2nd energy state. The energy released in this process, will be :

(Given Rch = $13.6\ \text{eV}$)

Where R = Rydberg constant

c = Speed of light in vacuum

h = Planck's constant

(A) $10.5\ \text{eV}$ (B) $40.8\ \text{eV}$ (C) $13.6\ \text{eV}$ (D) $3.4\ \text{eV}$

Open solution

Q74. JEE Main 2023 (Online) 1st February Morning Shift

The mass of proton, neutron and helium nucleus are respectively $1.0073\ u, 1.0087\ u$ and $4.0015\ u$. The binding energy of helium nucleus is :

(A) $28.4\ \text{MeV}$ (B) $56.8\ \text{MeV}$ (C) $7.1\ \text{MeV}$ (D) $14.2\ \text{MeV}$

Open solution

Q75. JEE Main 2023 (Online) 31st January Evening Shift

The radius of electron's second stationary orbit in Bohr's atom is R. The radius of 3rd orbit will be

(A) 2.25R (B) 3R (C) $\frac{\text{R}}{3}$ (D) 9R

Open solution

Q76. JEE Main 2023 (Online) 30th January Morning Shift

Speed of an electron in Bohr's 7th orbit for Hydrogen atom is $3.6 \times 10^6\ \text{m/s}$. The corresponding speed of the electron in 3rd orbit, in m/s is :

(A) (1.8×10^6) (B) (7.5×10^6) (C) (8.4×10^6) (D) (3.6×10^6)

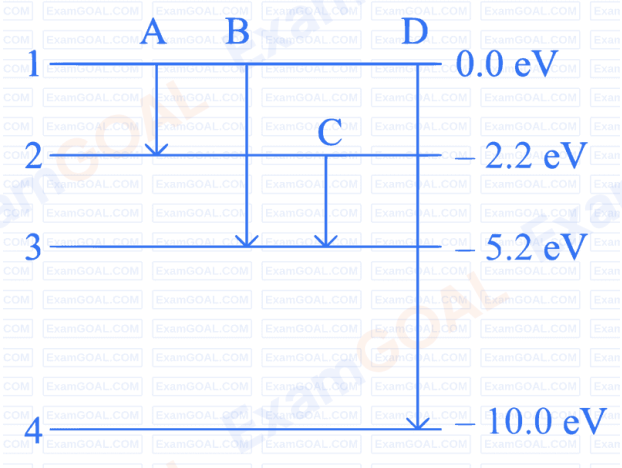
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Q77. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

The energy levels of an atom is shown in figure.



Which one of these transitions will result in the emission of a photon of wavelength 124.1 nm?

Given (h = 6.62 $\times$ 10^{-34} Js)

(A) C (B) B (C) A (D) D

Q78. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

The ratio of the density of oxygen nucleus ($^{16}_{8}O$) and helium nucleus ($^{4}_{2}\mathrm{He}$) is

(A) 4 : 1 (B) 1 : 1 (C) 2 : 1 (D) 8 : 1

Q79. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

A photon is emitted in transition from n = 4 to n = 1 level in hydrogen atom. The corresponding wavelength for this transition is (given, h = 4 $\times$ 10^{-15} eVs) :

(A) 99.3 nm (B) 94.1 nm (C) 974 nm (D) 941 nm

Q80. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

Using Boh'r model of quantization of angular momentum the relation between the radius 'r' of the nth allowed orbit of quantum number 'n' for an electron in hydrogen atom is:

(A) $r \propto n^{\frac{1}{2}}$ (B) $r \propto n^2$

(C) $r \propto \left(\frac{1}{n}\right)^2$ (D) $r \propto \left(\frac{1}{n}\right)^{\frac{1}{2}}$

Q81. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level.

(A) 3 : 4 (B) 4 : 3 (C) 1 : 4 (D) 4 : 1

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Q82. JEE Main 2022 (Online) 26th July Evening Shift

A nucleus of mass M at rest splits into two parts having masses $\frac{M'}{3}$ and $\frac{2M'}{3}(M' < M)$. The ratio of de Broglie wavelength of two parts will be :

(A) 1 : 2 (B) 2 : 1 (C) 1 : 1 (D) 2 : 3

Open solution

Q83. JEE Main 2022 (Online) 25th July Evening Shift

Hydrogen atom from excited state comes to the ground state by emitting a photon of wavelength λ. The value of principal quantum number 'n' of the excited state will be : (R : Rydberg constant)

(A) $\sqrt{\frac{\lambda \mathrm{R}}{\lambda-1}}$ (B) $\sqrt{\frac{\lambda \mathrm{R}}{\lambda \mathrm{R}-1}}$

(C) $\sqrt{\frac{\lambda}{\lambda \mathrm{R}-1}}$ (D) $\sqrt{\frac{\lambda R^2}{\lambda R-1}}$

Open solution

Q84. JEE Main 2022 (Online) 25th July Morning Shift

The momentum of an electron revolving in n^{th} orbit is given by :

(Symbols have their usual meanings)

(A) $\frac{\mathrm{nh}}{2\pi \mathrm{r}}$ (B) $\frac{nh}{2r}$

(C) $\frac{\mathrm{nh}}{2\pi}$ (D) $\frac{2\pi r}{\mathrm{nh}}$

Open solution

Q85. JEE Main 2022 (Online) 25th July Morning Shift

The magnetic moment of an electron (e) revolving in an orbit around nucleus with an orbital angular momentum is given by :

(A) $\vec{\mu}_{\mathrm{L}}=\frac{\overrightarrow{\mathrm{eL}}}{2\,\mathrm{m}}$ (B) $\vec{\mu}_{\mathrm{L}}=-\frac{\overrightarrow{\mathrm{eL}}}{2\,\mathrm{m}}$

(C) $\vec{\mu}_{l}=-\frac{\overrightarrow{eL}}{\mathrm{m}}$ (D) $\vec{\mu}_{l}=\frac{\overrightarrow{2\mathrm{eL}}}{\mathrm{m}}$

Open solution

Q86. JEE Main 2022 (Online) 30th June Morning Shift

A hydrogen atom in ground state absorbs 12.09 eV of energy. The orbital angular momentum of the electron is increased by :

(A) 1.05×10^{-34} Js (B) 2.11×10^{-34} Js

(C) 3.16×10^{-34} Js (D) 4.22×10^{-34} Js

Open solution

Q87. JEE Main 2022 (Online) 28th June Morning Shift

The Q-value of a nuclear reaction and kinetic energy of the projectile particle, K_p are related as :

(A) $Q = K_p$ (B) $(K_p + Q) < 0$ (C) $Q < K_p$ (D) $(K_p + Q) > 0$

Open solution

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Q88. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

Given below are two statements :

Statement I : In hydrogen atom, the frequency of radiation emitted when an electron jumps from lower energy orbit (E_1) to higher energy orbit (E_2), is given as $hf = E_1 - E_2$

Statement II : The jumping of electron from higher energy orbit (E_2) to lower energy orbit (E_1) is associated with frequency of radiation given as $f = (E_2 - E_1)/h$

This condition is Bohr's frequency condition.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is correct but Statement II is false.

(D) Statement I is incorrect but Statement II is true.

Q89. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

A hydrogen atom in its ground state absorbs 10.2 eV of energy. The angular momentum of electron of the hydrogen atom will increase by the value of :

(Given, Planck's constant = 6.6×10^{-34} Js).

(A) 2.10×10^{-34} Js

(B) 1.05×10^{-34} Js

(C) 3.15×10^{-34} Js

(D) 4.2×10^{-34} Js

Q90. JEE Main 2022 (Online) 25th June Morning Shift

Open solution

The ratio for the speed of the electron in the 3rd orbit of He^+ to the speed of the electron in the 3rd orbit of hydrogen atom will be :

(A) 1 : 1

(B) 1 : 2

(C) 4 : 1

(D) 2 : 1

Q91. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

In Bohr's atomic model of hydrogen, let K, P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level :

(A) All K, P and E increase.

(B) K decreases, P and E increase.

(C) P decreases, K and E increase.

(D) K increases, P and E decrease.

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Q92. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

Choose the correct option from the following options given below :

(A) In the ground state of Rutherford's model electrons are in stable equilibrium. While in Thomson's model electrons always experience a net-force.

(B) An atom has a nearly continuous mass distribution in a Rutherford's model but has a highly non-uniform mass distribution in Thomson's model.

(C) A classical atom based on Rutherford's model is doomed to collapse.

(D) The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model.

Q93. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 MeV. It splits into two fragments 'B' and 'C' of mass numbers 105 and 115. The binding energy of nucleons in 'B' and 'C' is 6.4 MeV per nucleon. The energy Q released per fission will be :

(A) 0.8 MeV (B) 275 MeV (C) 220 MeV (D) 176 MeV

Q94. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

A particular hydrogen like ion emits radiation of frequency 2.92×10^{15} Hz when it makes transition from n = 3 to n = 1. The frequency in Hz of radiation emitted in transition from n = 2 to n = 1 will be :

(A) 0.44×10^{15} (B) 6.57×10^{15} (C) 4.38×10^{15} (D) 2.46×10^{15}

Q95. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

A nucleus with mass number 184 initially at rest emits an α-particle. If the Q value of the reaction is 5.5 MeV, calculate the kinetic energy of the α-particle.

(A) 5.5 MeV (B) 5.0 MeV (C) 5.38 MeV (D) 0.12 MeV

Q96. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

A nucleus of mass M emits γ -ray photon of frequency 'v'. The loss of internal energy by the nucleus is :

[Take 'c' as the speed of electromagnetic wave]

(A) hv (B) $hv\left[1+\frac{hv}{2Mc^2}\right]$

(C) $hv\left[1-\frac{hv}{2Mc^2}\right]$ (D) 0

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Q97. JEE Main 2021 (Online) 18th March Morning Shift

Imagine that the electron in a hydrogen atom is replaced by a muon (μ). The mass of muon particle is 207 times that of an electron and charge is equal to the charge of an electron. The ionization potential of this hydrogen atom will be :

(A) 13.6 eV (B) 2815.2 eV (C) 331.2 eV (D) 27.2 eV

Open solution

Q98. JEE Main 2021 (Online) 17th March Evening Shift

The atomic hydrogen emits a line spectrum consisting of various series. Which series of hydrogen atomic spectra is lying in the visible region?

(A) Brackett series (B) Balmer series (C) Paschen series (D) Lyman series

Open solution

Q99. JEE Main 2021 (Online) 17th March Morning Shift

If an electron is moving in the nth orbit of the hydrogen atom, then its velocity (v_n) for the nth orbit is given as :

(A) $v_n \propto \frac{1}{n}$ (B) $v_n \propto n^2$ (C) $v_n \propto n$ (D) $v_n \propto \frac{1}{n^2}$

Open solution

Q100. JEE Main 2021 (Online) 17th March Morning Shift

Which level of the single ionized carbon has the same energy as the ground state energy of hydrogen atom?

(A) 8 (B) 6 (C) 1 (D) 4

Open solution

Q101. JEE Main 2021 (Online) 26th February Morning Shift

If λ_1 and λ_2 are the wavelengths of the third member of Lyman and first member of the Paschen series respectively, then the value of $\lambda_1 : \lambda_2$ is :

(A) 7 : 135 (B) 7 : 108 (C) 1 : 9 (D) 1 : 3

Open solution

Q102. JEE Main 2021 (Online) 25th February Evening Shift

The wavelength of the photon emitted by a hydrogen atom when an electron makes a transition from n = 2 to n = 1 state is :

(A) 194.8 nm (B) 490.7 nm (C) 913.3 nm (D) 121.8 nm

Open solution

Q103. JEE Main 2021 (Online) 24th February Evening Shift

According to Bohr atomic model, in which of the following transitions will the frequency be maximum?

(A) n = 2 to n = 1 (B) n = 3 to n = 2 (C) n = 4 to n = 3 (D) n = 5 to n = 4

Open solution

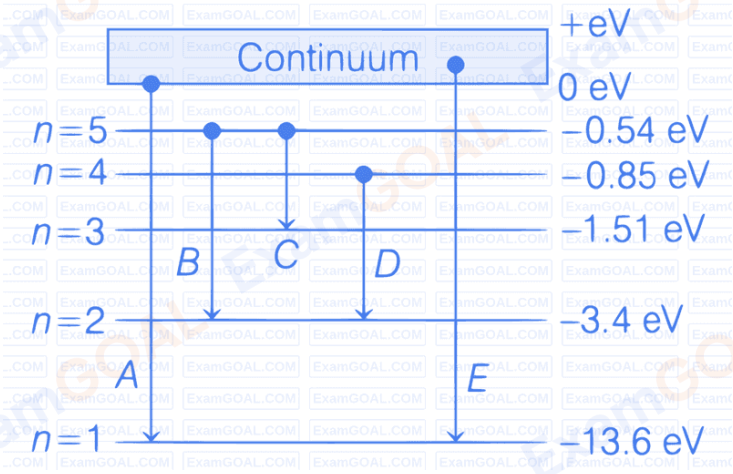
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Q104. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

In the given figure, the energy levels of hydrogen atom have been shown along with some transitions marked A, B, C, D and E.

The transitions A, B and C respectively represent :



(A) The series limit of Lyman series, second member of Balmer series and second member of Paschen series.

(B) The ionization potential of hydrogen, second member of Balmer series and third member of Paschen series.

(C) The series limit of Lyman series, third member of Balmer series and second member of Paschen series.

(D) The first member of the Lyman series, third member of Balmer series and second member of Paschen series.

Q105. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

Given the masses of various atomic particles
m_p = 1.0072 u, m_n = 1.0087 u, m_e = 0.000548 u,
$m_{\overline{v}}$ = 0, m_d = 2.0141 u, where p ≡ proton, n ≡ neutron,
e ≡ electron, $\overline{v}$ ≡ antineutrino and d ≡ deuteron. Which of the following process is allowed by momentum and energy conservation?

(A) $n + n \rightarrow$ deuterium atom
(electron bound to the nucleus)

(B) $n + p \rightarrow d + \gamma$

(C) $p \rightarrow n + e^+ + \overline{v}$

(D) $e^+ + e^- \rightarrow \gamma$

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Q106. JEE Main 2020 (Online) 6th September Morning Slot

You are given that Mass of $^{7}_{3}Li$ = 7.0160u,
Mass of $^{4}_{2}He$ = 4.0026u
and Mass of $^{1}_{1}H$ = 1.0079u.
When 20 g of $^{7}_{3}Li$ is converted into $^{4}_{2}He$ by proton capture, the energy liberated, (in kWh), is :
[Mass of nucleon = 1 GeV/c^2]

(A) 6.82×10^5 (B) 4.5×10^5 (C) 8×10^6 (D) 1.33×10^6

Open solution

Q107. JEE Main 2020 (Online) 4th September Evening Slot

Find the Binding energy per nucleon for $^{120}_{50}Sn$. Mass of proton m_p = 1.00783 U, mass of neutron m_n = 1.00867 U and mass of tin nucleus m_{Sn} = 119.902199 U. (take 1U = 931 MeV)

(A) 9.0 MeV (B) 8.5 MeV (C) 8.0 MeV (D) 7.5 MeV

Open solution

Q108. JEE Main 2020 (Online) 3rd September Evening Slot

The radius R of a nucleus of mass number A can be estimated by the formula
$R = (1.3 \times 10^{-15})A^{1/3}$ m.
It follows that the mass density of a nucleus is of the order of :

($M_{prot.} \cong M_{neut} \simeq 1.67 \times 10^{-27}$ kg)

(A) 10^{24} kg m^{-3} (B) 10^{10} kg m^{-3} (C) 10^{17} kg m^{-3} (D) 10^{3} kg m^{-3}

Open solution

Q109. JEE Main 2020 (Online) 3rd September Evening Slot

Hydrogen ion and singly ionized helium atom are accelerated, from rest, through the same potential difference. The ratio of final speeds of hydrogen and helium ions is close to :

(A) 2 : 1 (B) 1 : 2 (C) 5 : 7 (D) 10 : 7

Open solution

Q110. JEE Main 2020 (Online) 2nd September Evening Slot

In a hydrogen atom the electron makes a transition from $(n + 1)^{th}$ level to the nth level. If n >> 1, the frequency of radiation emitted is proportional to :

(A) $\frac{1}{n}$ (B) $\frac{1}{n^2}$ (C) $\frac{1}{n^3}$ (D) $\frac{1}{n^4}$

Open solution

Q111. JEE Main 2020 (Online) 2nd September Morning Slot

In a reactor, 2 kg of $_{92}U^{235}$ fuel is fully used up in 30 days. The energy released per fission is 200 MeV. Given that the Avogadro number, N = 6.023×10^{26} per kilo mole and 1 eV = 1.6×10^{-19} J. The power output of the reactor is close to

(A) 125 MW (B) 60 MW (C) 54 MW (D) 35 MW

Open solution

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Q112. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

The energy required to ionise a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state ?

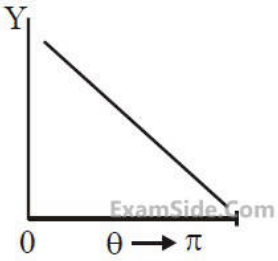
(A) 35.8 nm (B) 11.4 nm (C) 8.6 nm (D) 24.2 nm

Q113. JEE Main 2020 (Online) 8th January Morning Slot

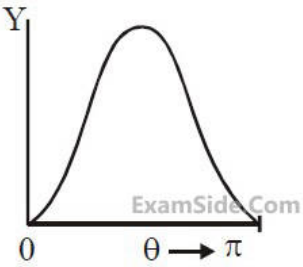
Open solution

The graph which depicts the results of Rutherford gold foil experiment with α-particales is :
θ : Scattering angle
Y : Number of scattered α-particles detected (Plots are schematic and not to scale)

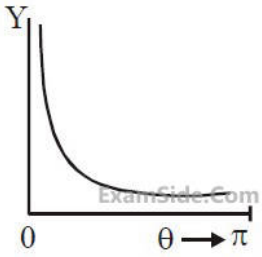
(A)



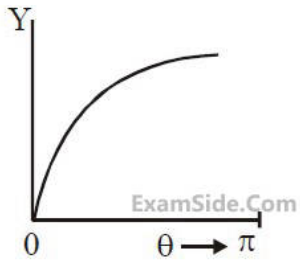
(B)



(C)



(D)



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Q114. JEE Main 2020 (Online) 7th January Morning Slot

The time period of revolution of electron in its ground state orbit in a hydrogen atom is 1.6×10^{-16} s. The frequency of revolution of the electron in its first excited state (in s^{-1}) is :

(A) 5.6×10^{12} (B) 1.6×10^{14} (C) 7.8×10^{14} (D) 6.2×10^{15}

Open solution

Q115. JEE Main 2019 (Online) 12th April Evening Slot

Consider an electron in a hydrogen atom revolving in its second excited state (having radius 4.65 $\overset{o}{A}$). The de-Broglie wavelength of this electron is :

(A) 6.6 $\overset{o}{A}$ (B) 3.5 $\overset{o}{A}$ (C) 9.7 $\overset{o}{A}$ (D) 12.9 $\overset{o}{A}$

Open solution

Q116. JEE Main 2019 (Online) 12th April Evening Slot

The electron in a hydrogen atom first jumps from the third excited state to the second excited state and subsequently to the first excited state. The ratio of the respective wavelengths, $\frac{\lambda_1}{\lambda_2}$, of the photons emitted in this process is :

(A) $\frac{22}{5}$ (B) $\frac{7}{5}$ (C) $\frac{9}{7}$ (D) $\frac{20}{7}$

Open solution

Q117. JEE Main 2019 (Online) 12th April Morning Slot

An excited He^+ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm, in making a transition to ground state. The quantum number n, corresponding to its initial excited state is (for photon of wavelength λ, energy $E = \frac{1240\ eV}{\lambda(in\ nm)}$) :

(A) n = 4 (B) n = 7 (C) n = 5 (D) n = 6

Open solution

Q118. JEE Main 2019 (Online) 10th April Evening Slot

In Li^{++}, electron in first Bohr orbit is excited to a level by a radiation of wavelength λ. When the ion gets deexcited to the ground state in all possible ways (including intermediate emissions), a total of six spectral lines are observed. What is the value of λ?
(Given : H = 6.63×10^{-34} Js; c = 3×10^8 ms $^{-1}$)

(A) 10.8 nm (B) 12.3 nm (C) 9.4 nm (D) 11.4 nm

Open solution

Q119. JEE Main 2019 (Online) 9th April Evening Slot

A He^+ ion is in its first excited state. Its ionization energy is :-

(A) 13.60 eV (B) 6.04 eV (C) 48.36 eV (D) 54.40 eV

Open solution

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Q120. JEE Main 2019 (Online) 9th April Morning Slot

Taking the wavelength of first Balmer line in hydrogen spectrum (n = 3 to n = 2) as 660 nm, the wavelength of the 2nd Balmer line (n = 4 to n = 2) will be :

(A) 642.7 nm (B) 488.9 nm (C) 889.2 nm (D) 388.9 nm

Open solution

Q121. JEE Main 2019 (Online) 8th April Evening Slot

The ratio of mass densities of nuclei of ^{40}Ca and ^{16}O is close to :-

(A) 1 (B) 5 (C) 0.1 (D) 2

Open solution

Q122. JEE Main 2019 (Online) 8th April Morning Slot

Radiation coming from transitions n = 2 to n = 1 of hydrogen atoms fall on He^+ ions in n = 1 and n = 2 states. The possible transition of helium ions as they absorb energy from the radiation is :

(A) n = 1 $\rightarrow$ n = 4 (B) n = 2 $\rightarrow$ n = 5 (C) n = 2 $\rightarrow$ n = 4 (D) n = 2 $\rightarrow$ n = 3

Open solution

Q123. JEE Main 2019 (Online) 12th January Morning Slot

A particle of mass m moves in a circular orbit in a central potential field U(r) = $\frac{1}{2}$ kr^2. If Bohr 's quantization conditions are applied, radii of possible orbitls and energy levels vary with quantum number n as :

(A) $r_n \propto \sqrt{n}$, $E_n \propto n$ (B) $r_n \propto \sqrt{n}$, $E_n \propto \frac{1}{n}$

(C) $r_n \propto n$, $E_n \propto n$ (D) $r_n \propto n^2$, $E_n \propto \frac{1}{n^2}$

Open solution

Q124. JEE Main 2019 (Online) 11th January Evening Slot

In a hydrogen like atom, when an electron jumps from the M-shell to the L-shell, the wavelength of emitted radiation is λ. If an electron jumps from N-shell to the L-shell, the wavelength of emitted radiation will be:

(A) $\frac{25}{16}\lambda$ (B) $\frac{27}{20}\lambda$ (C) $\frac{16}{25}\lambda$ (D) $\frac{20}{27}\lambda$

Open solution

Q125. JEE Main 2019 (Online) 11th January Morning Slot

A hydrogen atom, initially in the ground state is excited by absorbing a photon of wavelength 980$\mathring{A}$. The radius of the atom in the excited state, in terms of Bohr radius a_0 will be : (hc = 12500 eV$\mathring{A}$)

(A) $4a_0$ (B) $9a_0$ (C) $25a_0$ (D) $16a_0$

Open solution

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Q126. JEE Main 2019 (Online) 10th January Evening Slot

Consider the nuclear fission

$Ne^{20} \rightarrow 2He^4 + C^{12}$

Given that the binding energy/ nucleon of Ne^{20}, He^4 and C^{12} are, respectively, 8.03 MeV, 7.07 MeV and 7.86 MeV, identify the correct statement -

(A) 8.3 MeV energy will be released

(B) energy of 11.9 MeV has to be supplied

(C) energy of 12.4 MeV will be supplied

(D) energy of 3.6 MeV will be released

Open solution

Q127. JEE Main 2018 (Online) 16th April Morning Slot

Both the nucleus and the atom of some element are in their respective first excited states. They get de-excited by emitting photons of wavelengths λ_N, λ_A respectively. The ratio $\frac{\lambda_N}{\lambda_A}$ is closest to :

(A) 10^{-6} (B) 10 (C) 10^{-10} (D) 10^{-1}

Open solution

Q128. JEE Main 2018 (Offline)

An electron from various excited states of hydrogen atom emit radiation to come to the ground state. Let λ_n, λ_g be the de Broglie wavelength of the electron in the nth state and the ground state respectively. Let Λ_n be the wavelength of the emitted photon in the transition from the nth state to the ground state. For large n, (A, B are constants)

(A) $\Lambda_n \approx A + \frac{B}{\lambda_n^2}$

(B) $\Lambda_n \approx A + B\lambda_n$

(C) $\Lambda_n^2 \approx A + B\lambda_n^2$

(D) $\Lambda_n^2 \approx \lambda$

Open solution

Q129. JEE Main 2018 (Offline)

If the series limit frequency of the Lyman series is ν_L, then the series limit frequency of the Pfund series is:

(A) $\nu_L/25$ (B) $25\nu_L$ (C) $16\nu_L$ (D) $\nu_L/16$

Open solution

Q130. JEE Main 2018 (Online) 15th April Evening Slot

Muon (μ^-) is a negatively charged (|q| = |e|) particle with a mass m_μ = 200 m_e, where m_e is the mass of the electron and e is the electronic charge. If μ^- is bound to a proton to form a hydrogen like atom, identify the correct statements.

(A) Radius of the muonic orbit is 200 times smaller than that of the electron.

(B) The speed of the μ^- in the nth orbit is $\frac{1}{200}$ times that of the electron in the nth orbit.

(C) The ionization energy of muonic atom is 200 times more than that of a hydrogen atom.

(D) The momentum of the muon in the nth orbit is 200 times more than that of the electron.

(A) (A), (B), (D)

(B) (A), (C), (D)

(C) (B), (D)

(D) (C), (D)

Open solution

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Q131. JEE Main 2018 (Online) 15th April Evening Slot

An unstable heavy nucleus at rest breaks into two nuclei which move away with velocities in the ratio of 8 : 27. The ratio of the radii of the nuclei (assumed to be spherical) is :

(A) 8 : 27 (B) 4 : 9 (C) 3 : 2 (D) 2 : 3

Open solution

Q132. JEE Main 2018 (Online) 15th April Morning Slot

The energy required to remove the electron from a singly ionized Helium atom is 2.2 times the energies required to remove an electron from Helium atom. The total energy required to ionize the Helium atom completely is :

(A) $20\ eV$ (B) $34\ eV$ (C) $79\ eV$ (D) $109\ eV$

Open solution

Q133. JEE Main 2017 (Online) 9th April Morning Slot

The acceleration of an electron in the first orbit of the hydrogen atom (n = 1) is :

(A) $\frac{h^2}{\pi^2 m^2 r^3}$ (B) $\frac{h^2}{8\pi^2 m^2 r^3}$
(C) $\frac{h^2}{4\pi^2 m^2 r^3}$ (D) $\frac{h^2}{4\pi m^2 r^3}$

Open solution

Q134. JEE Main 2017 (Online) 9th April Morning Slot

Imagine that a reactor converts all given mass into energy and that it operates at a power level of 10^9 watt. The mass of the fuel consumed per hour in the reactor will be : (velocity of light, c is 3×10^8 m/s)

(A) 0.96 gm (B) 0.8 gm (C) 4×10^{-2} gm (D) 6.6×10^{-5} gm

Open solution

Q135. JEE Main 2017 (Online) 8th April Morning Slot

According to Bohr's theory, the time averaged magnetic field at the centre (i.e. nucleus) of a hydrogen atom due to the motion of electrons in the nth orbit is proportional to : (n = principal quantum number)

(A) n^{-4} (B) n^{-5} (C) n^{-3} (D) n^{-2}

Open solution

Q136. JEE Main 2017 (Online) 8th April Morning Slot

Two deuterons undergo nuclear fusion to form a Helium nucleus. Energy released in this process is : (given binding energy per nucleon for deuteron = 1.1 MeV and for helium = 7.0 MeV)

(A) 30.2 MeV (B) 32.4 MeV (C) 23.6 MeV (D) 25.8 MeV

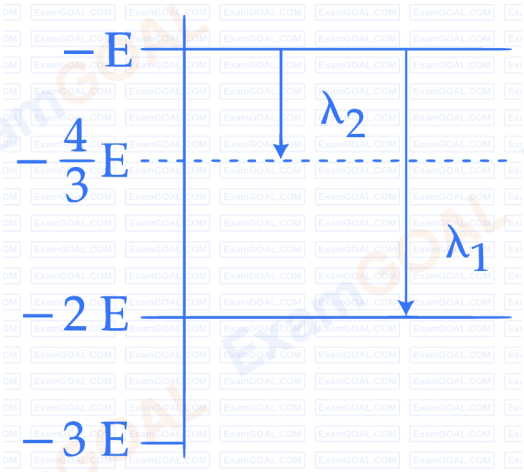
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Q137. JEE Main 2017 (Offline)

Some energy levels of a molecule are shown in the figure. The ratio of the wavelengths r = λ_1/λ_2, is given by:

Open solution



(A) r = 1/3 (B) r = 4/3 (C) r = 2/3 (D) r = 3/4

Q138. JEE Main 2016 (Online) 9th April Morning Slot

A hydrogen atom makes a transition from n = 2 to n = 1 and emits a photon. This photon strikes a doubly ionized lithium atom (z = 3) in excited state and completely removes the orbiting electron. The least quantum number for the excited state of the ion for the process is :

Open solution

(A) 2 (B) 3 (C) 4 (D) 5

Q139. JEE Main 2015 (Online) 11th April Morning Slot

The de - Broglie wavelength associated with the electron in the n = 4 level is :

Open solution

(A) two times the de-Broglie wavelength of the electron in the ground state

(B) four times the de -Broglie wavelength of the electron in the ground state

(C) half of the de-Broglie wavelength of the electron in the ground state

(D) 1/4th of the de-Broglie wavelength of the electron in the ground state

Q140. JEE Main 2015 (Online) 10th April Morning Slot

If one were to apply Bohr model to a particle of mass 'm' and charge 'q' moving in a plane under the influence of a magnetic field 'B', the energy of the charged particle in the nth level will be :

Open solution

(A) $n\left(\frac{hqB}{2\pi m}\right)$ (B) $n\left(\frac{hqB}{4\pi m}\right)$

(C) $n\left(\frac{hqB}{8\pi m}\right)$ (D) $n\left(\frac{hqB}{\pi m}\right)$

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Q141. JEE Main 2015 (Offline)

As an electron makes a transition from an excited state to the ground state of a hydrogen - like atom/ion :

(A) kinetic energy decreases, potential energy increases but total energy remains same

(B) kinetic energy and total energy decrease but potential energy increases

(C) its kinetic energy increases but potential energy and total energy decrease

(D) kinetic energy, potential energy and total energy decrease

Open solution

Q142. JEE Main 2014 (Online) 9th April Morning Slot

If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{++} is :

(A) 122.4 eV (B) 30.6 eV (C) 13.6 eV (D) 3.4 eV

Open solution

Q143. JEE Main 2014 (Offline)

The radiation corresponding to $3 \rightarrow 2$ transition of hydrogen atom falls on a metal surface to produce photoelectrons. These electrons are made to enter a magnetic field $3 \times 10^{-4}\,T$. If the radius of the larger circular path followed by these electrons is $10.0\ mm$, the work function of the metal is close to:

(A) $1.8\ eV$ (B) $1.1\ eV$ (C) $0.8\ eV$ (D) $1.6\ eV$

Open solution

Q144. JEE Main 2014 (Offline)

Hydrogen $\left({}_1H^1\right)$, Deuterium $\left({}_1H^2\right)$, singly ionised Helium $\left({}_2He^4\right)^+$ and doubly ionised lithium $\left({}_3Li^6\right)^{++}$ all have one electron around the nucleus. Consider an electron transition from $n = 2$ to $n = 1$. If the wavelengths of emitted radiation are $\lambda_1, \lambda_2, \lambda_3$ and λ_4 respectively then approximately which one of the following is correct?

(A) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$ (B) $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(C) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$ (D) $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$

Open solution

Q145. JEE Main 2013 (Online) 25th April Morning Slot

A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. It will emit:

(A) 2 lines in the Lyman series and 1 line in the Balmer series

(B) 3 lines in the Lyman series

(C) 1 line in the Lyman series and 2 lines in the Balmer series

(D) 3 lines in the Balmer series

Open solution

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Q146. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

In the Bohr's model of hydrogen-like atom the force between the nucleus and the electron is modified as $F = \frac{e^2}{4\pi\epsilon_0}\left(\frac{1}{r^2} + \frac{\beta}{r^3}\right)$, where β is a constant. For this atom, the radius of the nth orbit in terms of the Bohr radius $\left(a_0 = \frac{\epsilon_0 h^2}{m\pi e^2}\right)$ is :

(A) $r_n = a_0 n - \beta$ (B) $r_n = a_0 n^2 + \beta$ (C) $r_n = a_0 n^2 - \beta$ (D) $r_n = a_0 n + \beta$

Q147. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

In the Bohr model an electron moves in a circular orbit around the proton. Considering the orbiting electron to be a circular current loop, the magnetic moment of the hydrogen atom, when the electron is in n^{th} excited state, is :

(A) $\left(\frac{e}{2m}\right)\frac{n^2 h}{2\pi}$ (B) $\left(\frac{e}{m}\right)\frac{nh}{2\pi}$

(C) $\left(\frac{e}{2m}\right)\frac{nh}{2\pi}$ (D) $\left(\frac{e}{m}\right)\frac{n^2 h}{2\pi}$

Q148. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

When Uranium is bombarded with neutrons, it undergoes fission. The fission reaction can be written as :
${}_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{56}Ba^{141} + {}_{36}Kr^{92} + 3x + Q$(energy) where three particles named x are produced and energy Q is released. What is the name of the particle x?

(A) electron (B) α-particle (C) neutron (D) neutrino

Q149. JEE Main 2013 (Offline)

Open solution

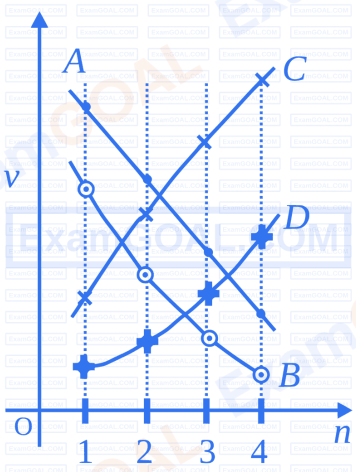
In a hydrogen like atom electron make transition from an energy level with quantum number n to another with quantum number $(n-1)$. If $n >> 1$, the frequency of radiation emitted is proportional to :

(A) $\frac{1}{n}$ (B) $\frac{1}{n^2}$ (C) $\frac{1}{n^{\frac{3}{2}}}$ (D) $\frac{1}{n^3}$

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Q150. AIEEE 2012 (Online) 26th May

Which of the plots shown in the figure represents speed (v) of the electron in a hydrogen atom as a function of the principal quantum number (n)?



Open solution

(A) B (B) D (C) C (D) A

Q151. AIEEE 2012 (Online) 19th May

Ionisation energy of Li (Lithium) atom in ground state is $5.4\,\text{eV}$. Binding energy of an electron in Li^+ ion in ground state is $75.6\,\text{eV}$. Energy required to remove all three electrons of Lithium (Li) atom is

(A) $81.0\,\text{eV}$ (B) $135.4\,\text{eV}$ (C) $203.4\,\text{eV}$ (D) $156.6\,\text{eV}$

Open solution

Q152. AIEEE 2012 (Online) 12th May

A hypothetical atom has only three energy levels. The ground level has energy, $E_1 = -8\,\text{eV}$. The two excited states have energies, $E_2 = -6\,\text{eV}$ and $E_3 = -2\,\text{eV}$. Then which of the following wavelengths will not be present in the emission spectrum of this atom?

(A) $207\,\text{nm}$ (B) $465\,\text{nm}$ (C) $310\,\text{nm}$ (D) $620\,\text{nm}$

Open solution

Q153. AIEEE 2012 (Online) 12th May

A doubly ionised Li atom is excited from its ground state $(n = 1)$ to $n = 3$ state. The wavelengths of the spectral lines are given by λ_{32}, λ_{31} and λ_{21}. The ratio $\lambda_{32}/\lambda_{31}$ and $\lambda_{21}/\lambda_{31}$ are, respectively

(A) $8.1, 0.67$ (B) $8.1, 1.2$ (C) $6.4, 1.2$ (D) $6.4, 0.67$

Open solution

Q154. AIEEE 2012 (Online) 7th May

The electron of a hydrogen atom makes a transition from the $(n+1)^{\text{th}}$ orbit to the nth orbit. For large n the wavelength of the emitted radiation is proportional to

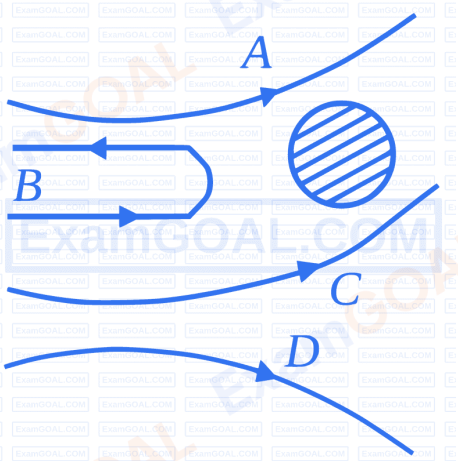
(A) n (B) n^3 (C) n^4 (D) n^2

Open solution

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Q155. AIEEE 2012 (Online) 7th May

In the Rutherford experiment, α-particles are scattered from a nucleus as shown. Out of the four paths, which path is not possible?



Open solution

(A) D (B) B (C) C (D) A

Q156. AIEEE 2012

Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. Then the number of spectral lines in the emission spectra will be :

(A) 2 (B) 3 (C) 5 (D) 6

Open solution

Q157. AIEEE 2012

Assume that a neutron breaks into a proton and an electron. The energy released during this process is : (mass of neutron $= 1.6725 \times 10^{-27} kg$, mass of proton $= 1.6725 \times 10^{-27}\, kg$, mass of electron $= 9 \times 10^{-31}\, kg$).

(A) $0.51\ MeV$ (B) $7.10\ MeV$ (C) $6.30\ MeV$ (D) $5.4\ MeV$

Open solution

Q158. AIEEE 2012

A diatomic molecule is made of two masses m_1 and m_2 which are separated by a distance r. If we calculate its rotational energy by applying Bohr's rule of angular momentum quantization, its energy will be given by: (n is an integer)

(A) $\frac{(m_1+m_2)^2 n^2 h^2}{2m_1^2 m_2^2 r^2}$ (B) $\frac{n^2h^2}{2(m_1+m_2)r^2}$

(C) $\frac{2n^2h^2}{(m_1+m_2)r^2}$ (D) $\frac{(m_1+m_2)n^2h^2}{2m_1m_2r^2}$

Open solution

Q159. AIEEE 2011

Energy required for the electron excitation in Li^{++} from the first to the third Bohr orbit is :

(A) $36.3\ eV$ (B) $108.8\ eV$ (C) $122.4\ eV$ (D) $12.1\ eV$

Open solution

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Q160. AIEEE 2010

A nucleus of mass $M+\Delta m$ is at rest and decays into two daughter nuclei of equal mass $\frac{M}{2}$ each. Speed of light is c.

The binding energy per nucleon for the parent nucleus is E_1 and that for the daughter nuclei is E_2. Then

(A) $E_2 = 2E_1$ (B) $E_1 > E_2$ (C) $E_2 > E_1$ (D) $E_1 = 2E_2$

Open solution

Q161. AIEEE 2010

A nucleus of mass $M+\Delta m$ is at rest and decays into two daughter nuclei of equal mass $\frac{M}{2}$ each. Speed of light is c.

The speed of daughter nuclei is

(A) $c\frac{\Delta m}{M+\Delta m}$ (B) $c\sqrt{\frac{2\Delta m}{M}}$

(C) $c\sqrt{\frac{\Delta m}{M}}$ (D) $c\sqrt{\frac{\Delta m}{M+\Delta m}}$

Open solution

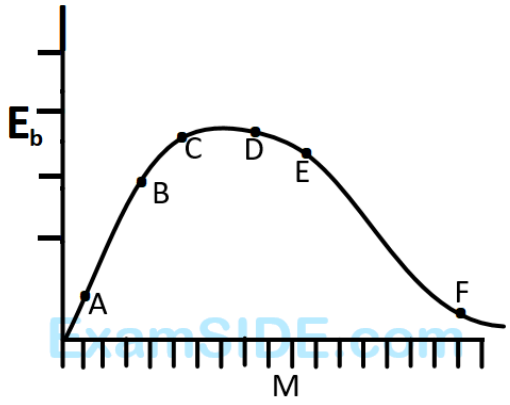
Q162. AIEEE 2009

The transition from the state $n = 4$ to $n = 3$ in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition from :

(A) $3 \to 2$ (B) $4 \to 2$ (C) $5 \to 4$ (D) $2 \to 1$

Open solution

Q163. AIEEE 2009



The above is a plot of binding energy per nucleon E_b, against the nuclear mass M; A, B, C, D, E, F correspond to different nuclei. Consider four reactions :

(i) $A + B \to C + \varepsilon$ (ii) $C \to A + B + \varepsilon$

(iii) $D + E \to F + \varepsilon$ (iv) $F \to D + E + \varepsilon$,

where ε is the energy released? In which reactions is ε positive?

(A) (i) and (iii) (B) (ii) and (iv) (C) (ii) and (iii) (D) (i) and (iv)

Open solution

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Q164. AIEEE 2008

This question contains Statement- 1 and Statement- 2. Of the four choices given after the statements, choose the one that best describes the two statements:

Statement- 1:

Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion.

Statement- 2:

For heavy nuclei, binding energy per nucleon increases with increasing Z while for light nuclei it decreases with increasing Z.

(A) Statement - 1 is false, Statement - 2 is true

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1

(C) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement - 1

(D) Statement - 1 is true, Statement - 2 is false

Open solution

Q165. AIEEE 2008

Suppose an electron is attracted towards the origin by a force $\frac{k}{r}$ where $'k'$ is a constant and $'r'$ is the distance of the electron from the origin. By applying Bohr model to this system, the radius of the nth orbital of the electron is found to be $'r_n'$ and the kinetic energy of the electron to be $'T_n'$. Then which of the following is true?

(A) $T_n \propto \frac{1}{n^2}, r_n \propto n^2$

(B) T_n independent of $n, r_n \propto n$

(C) $T_n \propto \frac{1}{n}, r_n \propto n$

(D) $T_n \propto \frac{1}{n}, r_n \propto n^2$

Open solution

Q166. AIEEE 2007

If M_O is the mass of an oxygen isotope ${}_8O^{17}$, M_p and M_N are the masses of a proton and neutron respectively, the nuclear binding energy of the isotope is

(A) $(M_O - 17M_N)C^2$

(B) $(M_O - 8M_P)C^2$

(C) $(M_O - 8M_P - 9M_N)C^2$

(D) $M_O c^2$

Open solution

Q167. AIEEE 2007

Which of the following transitions in hydrogen atoms emit photons of highest frequency ?

(A) $n = 1$ to $n = 2$

(B) $n = 2$ to $n = 6$

(C) $n = 6$ to $n = 2$

(D) $n = 2$ to $n = 1$

Open solution

Q168. AIEEE 2006

An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to

(A) v^2

(B) $\frac{1}{m}$

(C) $\frac{1}{v^4}$

(D) $\frac{1}{Ze}$

Open solution

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Q169. AIEEE 2006

When ${}_3Li^7$ nuclei are bombarded by protons, and the resultant nuclei are ${}_4Be^8$, the emitted particles will be

(A) alpha particles (B) beta particles (C) gamma photons (D) neutrons

Open solution

Q170. AIEEE 2006

If the binding energy per nucleon in ${}^7_3Li$ and ${}^4_2He$ nuclei are $5.60\ MeV$ and $7.06\ MeV$ respectively, then in the reaction

$$p + {}^7_3Li \rightarrow 2\,{}^4_2He$$

energy of proton must be

(A) $28.24\ MeV$ (B) $17.28\ MeV$ (C) $1.46\ MeV$ (D) $39.2\ MeV$

Open solution

Q171. AIEEE 2005

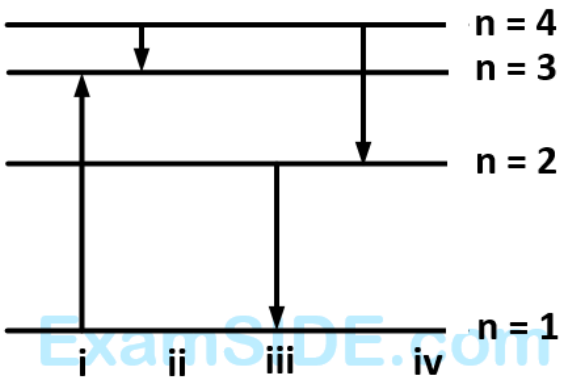
A nuclear transformation is denoted by $X\,(n,\alpha){}^7_3Li$. Which of the following is the nucleus of element X ?

(A) ${}^{10}_5B$ (B) ${}^{12}C_6$ (C) ${}^{11}_4Be$ (D) ${}^9_5B$

Open solution

Q172. AIEEE 2005

The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy?



(A) iv (B) iii (C) ii (D) i

Open solution

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Q173. AIEEE 2004

The binding energy per nucleon of deuteron $\left(_1^2H\right)$ and helium nucleus $\left(_2^4He\right)$ is $1.1\ MeV$ and $7\ MeV$ respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is

(A) $23.6\ MeV$ (B) $26.9\ MeV$ (C) $13.9\ MeV$ (D) $19.2\ MeV$

Open solution

Q174. AIEEE 2004

An α-particle of energy $5\ MeV$ is scattered through 180° by a fixed uranium nucleus. The distance of closest approach is of the order of

(A) $10^{-12}\ cm$ (B) $10^{-10}\ cm$ (C) $1A$ (D) $10^{-15a}\ cm$

Open solution

Q175. AIEEE 2003

In the nuclear fusion reaction $_1^2H + _1^3H \rightarrow _2^4He + n$
given that the repulsive potential energy between the two nuclei is $\sim 7.7 \times 10^{-14}J$, the temperature at which the gases must be heated to initiate the reaction is nearly
[Boltzmann's Constant $k = 1.38 \times 10^{-23}\ J/K$]

(A) $10^7\ K$ (B) $10^5\ K$ (C) $10^3\ K$ (D) $10^9\ K$

Open solution

Q176. AIEEE 2003

Which of the following atoms has the lowest ionization potential ?

(A) $_7^{14}N$ (B) $_{55}^{133}Cs$ (C) $_{18}^{40}Ar$ (D) $_8^{16}O$

Open solution

Q177. AIEEE 2003

The wavelengths involved in the spectrum of deuterium $\left(_1^2D\right)$ are slightly different from that of hydrogen spectrum, because

(A) the size of the two nuclei are different

(B) the nuclear forces are different in the two cases

(C) the masses of the two nuclei are different

(D) the attraction between the electron and the nucleus is different in the two cases

Open solution

Q178. AIEEE 2003

If the binding energy of the electron in a hydrogen atom is $13.6eV$, the energy required to remove the electron from the first excited state of Li^{++} is

(A) $30.6\ eV$ (B) $13.6\ eV$ (C) $3.4\ eV$ (D) $122.4\ eV$

Open solution

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Q179. AIEEE 2002

If mass-energy equivalence is taken into account, when water is cooled to form ice, the mass of water should

(A) increase
(B) remain unchanged
(C) decrease
(D) first increase then decrease

Open solution

Q180. AIEEE 2002

If $13.6\ eV$ energy is required to ionize the hydrogen atom, then the energy required to remove an electron from $n = 2$ is

(A) $10.2\ eV$ (B) $0\ eV$ (C) $3.4\ eV$ (D) $6.8\ eV$.

Open solution

Numerical 52 questions

Q181. JEE Main 2026 (Online) 6th April Morning Shift

The energy released when $\frac{7}{17.13}$ kg of $^{7}_{3}\mathrm{Li}$ is converted into $^{4}_{2}\mathrm{He}$ by proton bombardment is $\alpha \times 10^{32}\mathrm{eV}$. The value of α is ____ . (Nearest integer) (Mass of $^{7}_{3}\mathrm{Li} = 7.0183\mathrm{u}$, mass of $^{4}_{2}\mathrm{He} = 4.004\mathrm{u}$, mass of proton $= 1.008\mathrm{u}$ and $1\mathrm{u} = 931\mathrm{MeV/c^2}$ and Avogadro number $= 6.0 \times 10^{23}$)

Open solution

Q182. JEE Main 2026 (Online) 2nd April Evening Shift

Using Bohr's model, calculate the ratio of the magnetic fields generated due to the motion of the electrons in the 2nd and 4th orbits of hydrogen atom _____.

Open solution

Q183. JEE Main 2026 (Online) 23rd January Evening Shift

The average energy released per fission for the nucleus of $^{235}_{92}\mathrm{U}$ is 190 MeV . When all the atoms of 47 g pure $^{235}_{92}\mathrm{U}$ undergo fission process, the energy released is $\alpha \times 10^{23}\mathrm{MeV}$. The value of α is ____ . (Avogadro Number $= 6 \times 10^{23}$ per mole)

Open solution

Q184. JEE Main 2025 (Online) 3rd April Evening Shift

An electron in the hydrogen atom initially in the fourth excited state makes a transition to $\mathrm{n^{th}}$ energy state by emitting a photon of energy 2.86 eV . The integer value of n will be_____.

Open solution

Q185. JEE Main 2024 (Online) 9th April Morning Shift

A star has 100% helium composition. It starts to convert three $^{4}\mathrm{He}$ into one $^{12}\mathrm{C}$ via triple alpha process as $^{4}\mathrm{He} + {}^{4}\mathrm{He} + {}^{4}\mathrm{He} \rightarrow {}^{12}\mathrm{C} + \mathrm{Q}$. The mass of the star is 2.0×10^{32} kg and it generates energy at the rate of 5.808×10^{30} W. The rate of converting these $^{4}\mathrm{He}$ to $^{12}\mathrm{C}$ is $\mathrm{n} \times 10^{42}\ \mathrm{s^{-1}}$, where n is _____. [Take, mass of $^{4}\mathrm{He} = 4.0026\mathrm{u}$, mass of $^{12}\mathrm{C} = 12\mathrm{u}$]

Open solution

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Q186. JEE Main 2024 (Online) 8th April Morning Shift

In an alpha particle scattering experiment distance of closest approach for the α particle is $4.5 \times 10^{-14}\ \mathrm{m}$. If target nucleus has atomic number 80 , then maximum velocity of α-particle is _____ $\times 10^5\ \mathrm{m/s}$ approximately.

($\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \mathrm{SI}$ unit, mass of α particle $= 6.72 \times 10^{-27}\ \mathrm{kg}$)

Open solution

Q187. JEE Main 2024 (Online) 6th April Morning Shift

Radius of a certain orbit of hydrogen atom is 8.48 $\overset{o}{A}$. If energy of electron in this orbit is E/x. then $x =$ _____ (Given $\mathrm{a_0} = 0.529\ \overset{o}{A}$, $E =$ energy of electron in ground state).

Open solution

Q188. JEE Main 2024 (Online) 5th April Evening Shift

The shortest wavelength of the spectral lines in the Lyman series of hydrogen spectrum is $915\overset{o}{A}$. The longest wavelength of spectral lines in the Balmer series will be _____ $\overset{o}{A}$.

Open solution

Q189. JEE Main 2024 (Online) 5th April Morning Shift

If three helium nuclei combine to form a carbon nucleus then the energy released in this reaction is _____ $\times 10^{-2}\ \mathrm{MeV}$. (Given $1\mathrm{u} = 931\ \mathrm{MeV/c^2}$, atomic mass of helium $= 4.002603\mathrm{u}$)

Open solution

Q190. JEE Main 2024 (Online) 4th April Evening Shift

The disintegration energy Q for the nuclear fission of $^{235}\mathrm{U} \rightarrow {}^{140}\mathrm{Ce} + {}^{94}\mathrm{Zr} + n$ is _____ MeV.

Given atomic masses of $^{235}\mathrm{U} : 235.0439u$; $^{140}\mathrm{Ce} : 139.9054u$, $^{94}\mathrm{Zr} : 93.9063u$; $n : 1.0086u$, Value of $c^2 = 931\ \mathrm{MeV/u}$.

Open solution

Q191. JEE Main 2024 (Online) 4th April Morning Shift

A hydrogen atom changes its state from $n = 3$ to $n = 2$. Due to recoil, the percentage change in the wavelength of emitted light is approximately 1×10^{-n}. The value of n is _____.

[Given Rhc $= 13.6\ \mathrm{eV}, \mathrm{hc} = 1242\ \mathrm{eV\ nm}, \mathrm{h} = 6.6 \times 10^{-34}\ \mathrm{J\ s}$ mass of the hydrogen atom $= 1.6 \times 10^{-27}\ \mathrm{kg}$]

Open solution

Q192. JEE Main 2024 (Online) 1st February Evening Shift

A particular hydrogen-like ion emits the radiation of frequency $3 \times 10^{15}\ \mathrm{Hz}$ when it makes transition from $n = 2$ to $n = 1$. The frequency of radiation emitted in transition from $n = 3$ to $n = 1$ is $\frac{x}{9} \times 10^{15}\ \mathrm{Hz}$, when $x =$ _____ .

Open solution

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Q193. JEE Main 2024 (Online) 1st February Morning Shift

The radius of a nucleus of mass number 64 is 4.8 fermi. Then the mass number of another nucleus having radius of 4 fermi is $\frac{1000}{x}$, where x is _____.

Open solution

Q194. JEE Main 2024 (Online) 31st January Evening Shift

A nucleus has mass number A_1 and volume V_1. Another nucleus has mass number A_2 and Volume V_2. If relation between mass number is $A_2 = 4A_1$, then $\frac{V_2}{V_1} =$ _____.

Open solution

Q195. JEE Main 2024 (Online) 31st January Morning Shift

The mass defect in a particular reaction is $0.4\,\mathrm{g}$. The amount of energy liberated is $n \times 10^7\,\mathrm{kWh}$, where $n =$ _____. (speed of light $= 3 \times 10^8\,\mathrm{m/s}$)

Open solution

Q196. JEE Main 2024 (Online) 30th January Morning Shift

An electron of hydrogen atom in an excited state is having energy $\mathrm{E_n} = -0.85\,\mathrm{eV}$. The maximum number of allowed transitions to lower energy level is _____.

Open solution

Q197. JEE Main 2024 (Online) 29th January Evening Shift

Hydrogen atom is bombarded with electrons accelerated through a potential difference of V, which causes excitation of hydrogen atoms. If the experiment is being performed at $\mathrm{T} = 0\,\mathrm{K}$, the minimum potential difference needed to observe any Balmer series lines in the emission spectra will be $\frac{\alpha}{10}\,\mathrm{V}$, where $\alpha =$ _____.

Open solution

Q198. JEE Main 2024 (Online) 29th January Morning Shift

When a hydrogen atom going from $n = 2$ to $n = 1$ emits a photon, its recoil speed is $\frac{x}{5}\,\mathrm{m/s}$. Where $x =$ _____. (Use, mass of hydrogen atom $= 1.6 \times 10^{-27}\,\mathrm{kg}$)

Open solution

Q199. JEE Main 2024 (Online) 27th January Evening Shift

If Rydberg's constant is R, the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where $\alpha =$ _____.

Open solution

Q200. JEE Main 2024 (Online) 27th January Morning Shift

In a nuclear fission process, a high mass nuclide $(A \approx 236)$ with binding energy $7.6\,\mathrm{MeV}/$ Nucleon dissociated into middle mass nuclides $(\mathrm{A} \approx 118)$, having binding energy of $8.6\,\mathrm{MeV/Nucleon}$. The energy released in the process would be _____ MeV.

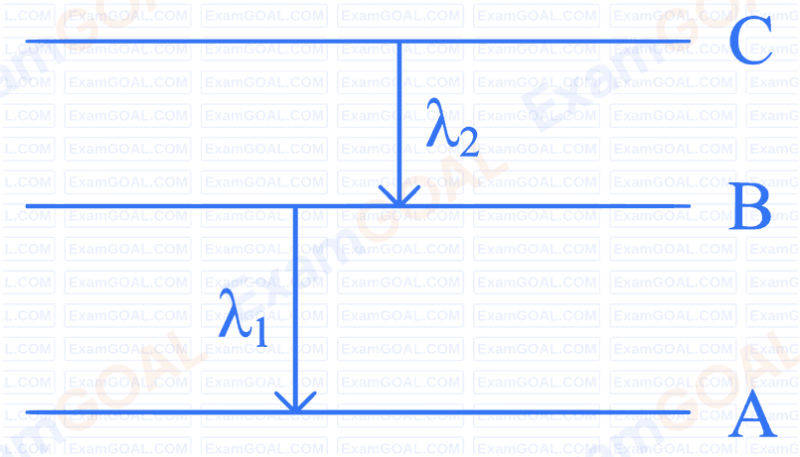
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Q201. JEE Main 2023 (Online) 15th April Morning Shift

As per given figure A, B and C are the first, second and third excited energy levels of hydrogen atom respectively. If the ratio of the two wavelengths (i.e. $\left(\frac{\lambda_1}{\lambda_2}\right)$ is $\frac{7}{4n}$, then the value of n will be _____.

Open solution



Q202. JEE Main 2023 (Online) 13th April Morning Shift

The radius of 2nd orbit of He^+ of Bohr's model is r_1 and that of fourth orbit of Be^{3+} is represented as r_2. Now the ratio $\frac{r_2}{r_1}$ is $x:1$. The value of x is _____.

Open solution

Q203. JEE Main 2023 (Online) 12th April Morning Shift

A common example of alpha decay is $^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}_2He^4 + Q$

Given :

$^{238}_{92}U = 238.05060\ u$,

$^{234}_{90}Th = 234.04360\ u$,

$^{4}_{2}He = 4.00260\ u$ and

$1u = 931.5\frac{MeV}{c^2}$

The energy released (Q) during the alpha decay of $^{238}_{92}U$ is _____ MeV

Open solution

Q204. JEE Main 2023 (Online) 11th April Evening Shift

A nucleus disintegrates into two nuclear parts, in such a way that ratio of their nuclear sizes is $1:2^{1/3}$. Their respective speeds have a ratio of $n:1$. The value of n is _____.

Open solution

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Q205. JEE Main 2023 (Online) 11th April Morning Shift

A monochromatic light is incident on a hydrogen sample in ground state. Hydrogen atoms absorb a fraction of light and subsequently emit radiation of six different wavelengths. The frequency of incident light is $x \times 10^{15}$ Hz. The value of x is _____.

(Given h $= 4.25 \times 10^{-15}$ eVs)

Open solution

Q206. JEE Main 2023 (Online) 10th April Evening Shift

If 917 $\overset{o}{A}$ be the lowest wavelength of Lyman series then the lowest wavelength of Balmer series will be _____ $\overset{o}{A}$.

Open solution

Q207. JEE Main 2023 (Online) 8th April Evening Shift

The ratio of wavelength of spectral lines H_α and H_β in the Balmer series is $\frac{x}{20}$. The value of x is _____.

Open solution

Q208. JEE Main 2023 (Online) 8th April Morning Shift

A nucleus with mass number 242 and binding energy per nucleon as 7.6 MeV breaks into two fragment each with mass number 121. If each fragment nucleus has binding energy per nucleon as 8.1 MeV, the total gain in binding energy is _____ MeV.

Open solution

Q209. JEE Main 2023 (Online) 6th April Evening Shift

Experimentally it is found that 12.8 eV energy is required to separate a hydrogen atom into a proton and an electron. So the orbital radius of the electron in a hydrogen atom is $\frac{9}{x} \times 10^{-10}$ m. The value of the x is _____.

$\left(1\text{eV} = 1.6 \times 10^{-19}\text{ J}, \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{Nm}^2/\text{C}^2 \text{ and electronic charge} = 1.6 \times 10^{-19}\text{C}\right)$

Open solution

Q210. JEE Main 2023 (Online) 6th April Morning Shift

The radius of fifth orbit of the Li^{++} is _____ $\times 10^{-12}$ m.

Take: radius of hydrogen atom $= 0.51\ \overset{o}{A}$

Open solution

Q211. JEE Main 2023 (Online) 1st February Evening Shift

Nucleus A having $Z = 17$ and equal number of protons and neutrons has 1.2 MeV binding energy per nucleon.

Another nucleus B of $Z = 12$ has total 26 nucleons and 1.8 MeV binding energy per nucleons.

The difference of binding energy of B and A will be _____ MeV.

Open solution

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Q212. JEE Main 2023 (Online) 1st February Morning Shift

A light of energy $12.75\ \text{eV}$ is incident on a hydrogen atom in its ground state. The atom absorbs the radiation and reaches to one of its excited states. The angular momentum of the atom in the excited state is $\frac{x}{\pi} \times 10^{-17}\ \text{eVs}$. The value of x is _____ (use $h = 4.14 \times 10^{-15}\ \text{eVs}, c = 3 \times 10^{8}\ \text{ms}^{-1}$).

Open solution

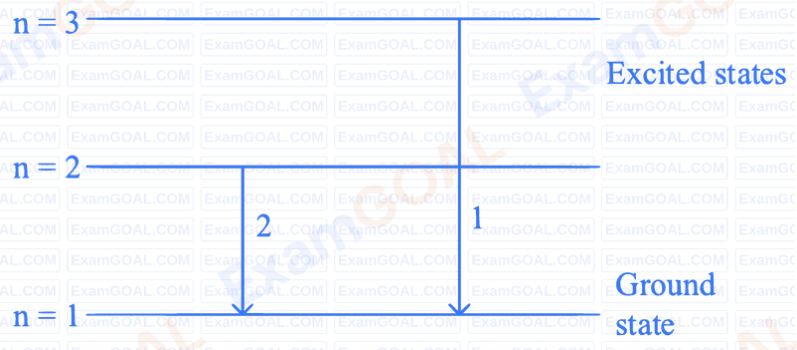
Q213. JEE Main 2023 (Online) 31st January Evening Shift

If the binding energy of ground state electron in a hydrogen atom is $13.6\,\text{eV}$, then, the energy required to remove the electron from the second excited state of Li^{2+} will be : $x \times 10^{-1}\text{eV}$. The value of x is _____.

Open solution

Q214. JEE Main 2023 (Online) 31st January Morning Shift

For hydrogen atom, λ_1 and λ_2 are the wavelengths corresponding to the transitions 1 and 2 respectively as shown in figure. The ratio of λ_1 and λ_2 is $\frac{x}{32}$. The value of x is _____.



Open solution

Q215. JEE Main 2023 (Online) 25th January Morning Shift

The wavelength of the radiation emitted is λ_0 when an electron jumps from the second excited state to the first excited state of hydrogen atom. If the electron jumps from the third excited state to the second orbit of the hydrogen atom, the wavelength of the radiation emitted will be $\frac{20}{x}\lambda_0$. The value of x is _____.

Open solution

Q216. JEE Main 2023 (Online) 24th January Evening Shift

The energy released per fission of nucleus of ^{240}X is 200 MeV. The energy released if all the atoms in 120g of pure ^{240}X undergo fission is _____ $\times 10^{25}$ MeV.

(Given $\text{N}_\text{A} = 6 \times 10^{23}$)

Open solution

Q217. JEE Main 2022 (Online) 30th July Evening Shift

A nuclear physicist performed an experiment on the scattering of alpha particles by thin metal foil of gold. The alpha particles were accelerated up to kinetic energy 3.2 Mev. The estimation of the radius of the nucleus of gold element ($Z = 79$) by him is _____ $\times 10^{-16}\ \text{m}$

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^{9}\ \frac{\text{N m}^2}{\text{C}^2}$)

Open solution

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Q218. JEE Main 2022 (Online) 26th July Evening Shift

Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below :

$_1^2X + _1^2X = _2^4Y$

The binding energies per nucleon for $_1^2X$ and $_2^4Y$ are $1.1\,\mathrm{MeV}$ and $7.6\,\mathrm{MeV}$ respectively. The energy released in this process is _____ MeV.

Open solution

Q219. JEE Main 2022 (Online) 26th July Morning Shift

In the hydrogen spectrum, λ be the wavelength of first transition line of Lyman series. The wavelength difference will be "aλ" between the wavelength of 3rd transition line of Paschen series and that of 2nd transition line of Balmer series where $a =$ _____.

Open solution

Q220. JEE Main 2022 (Online) 25th July Evening Shift

$\frac{x}{x+4}$ is the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its

(i) third permitted energy level to the second level and

(ii) the highest permitted energy level to the second permitted level.

The value of x will be _____.

Open solution

Q221. JEE Main 2022 (Online) 30th June Morning Shift

A hydrogen atom in its first excited state absorbs a photon of energy x $\times$ 10^{-2} eV and excited to a higher energy state where the potential energy of electron is -1.08 eV. The value of x is _____.

Open solution

Q222. JEE Main 2022 (Online) 29th June Morning Shift

$\sqrt{d_1}$ and $\sqrt{d_2}$ are the impact parameters corresponding to scattering angles 60° and 90° respectively, when an α particle is approaching a gold nucleus. For d_1 = x d_2, the value of x will be _____.

Open solution

Q223. JEE Main 2022 (Online) 27th June Morning Shift

A beam of monochromatic light is used to excite the electron in Li^{++} from the first orbit to the third orbit. The wavelength of monochromatic light is found to be x $\times$ 10^{-10} m. The value of x is _____.

[Given hc = 1242 eV nm]

Open solution

Q224. JEE Main 2021 (Online) 27th August Evening Shift

X different wavelengths may be observed in the spectrum from a hydrogen sample if the atoms are excited to states with principal quantum number n = 6 ? The value of X is _____.

Open solution

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Q225. JEE Main 2021 (Online) 27th July Evening Shift

The K_α X-ray of molybdenum has wavelength 0.071 nm. If the energy of a molybdenum atom with a K electron knocked out is 27.5 keV, the energy of this atom when an L electron is knocked out will be _____ keV. (Round off to the nearest integer)

[h = 4.14 $\times$ 10^{-15} eVs, c = 3 $\times$ 10^{8} ms^{-1}]

Open solution

Q226. JEE Main 2021 (Online) 27th July Morning Shift

In Bohr's atomic model, the electron is assumed to revolve in a circular orbit of radius 0.5 $\overset{o}{A}$. If the speed of electron is 2.2 $\times$ 10^{6} m/s, then the current associated with the electron will be _____ $\times$ 10^{-2} mA. [Take π as $\frac{22}{7}$]

Open solution

Q227. JEE Main 2021 (Online) 25th July Evening Shift

From the given data, the amount of energy required to break the nucleus of aluminium $^{27}_{13}Al$ is _____ x $\times$ 10^{-3} J.

Mass of neutron = 1.00866 u

Mass of proton = 1.00726 u

Mass of Aluminium nucleus = 27.18846 u

(Assume 1 u corresponds to x J of energy)

(Round off to the nearest integer)

Open solution

Q228. JEE Main 2021 (Online) 17th March Evening Shift

A particle of mass m moves in a circular orbit in a central potential field $U(r) = U_0 r^4$. If Bohr's quantization conditions are applied, radii of possible orbitals r_n vary with $n^{\frac{1}{\alpha}}$, where α is _____.

Open solution

Q229. JEE Main 2021 (Online) 16th March Morning Shift

The first three spectral lines of H-atom in the Balmer series are given λ_1, λ_2, λ_3 considering the Bohr atomic model, the wave lengths of first and third spectral lines $\left(\frac{\lambda_1}{\lambda_3}\right)$ are related by a factor of approximately 'x' $\times$ 10^{-1}.

The value of x, to the nearest integer, is _____.

Open solution

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Q230. JEE Main 2020 (Online) 5th September Morning Slot

A particle of mass 200 MeV/c^2 collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is
$\frac{N}{4}$. The value of N is :
(Given the mass of the hydrogen atom to be 1 GeV/c^2) _____ .

Open solution

Q231. JEE Main 2020 (Online) 4th September Morning Slot

In the line spectra of hydrogen atoms, difference between the largest and the shortest wavelengths of the Lyman series is 304 $\overset{0}{A}$. The corresponding difference for the Paschan series in $\overset{0}{A}$ is :
_____.

Open solution

Q232. JEE Main 2020 (Online) 8th January Evening Slot

The first member of the Balmer series of hydrogen atom has a wavelength of 6561 Å. The wavelength of the second member of the Balmer series (in nm) is:

Open solution

Out of syllabus 92 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q233. JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

The half-life of a radioactive nucleus is 5 years. The fraction of the original sample that would decay in 15 years is:

(A) $\frac{1}{8}$ (B) $\frac{3}{4}$ (C) $\frac{7}{8}$ (D) $\frac{1}{4}$

Open solution

Q234. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : The binding energy per nucleon is practically independent of the atomic number for nuclei of mass number in the range 30 to 170 .

Reason R : Nuclear force is short ranged.

In the light of the above statements, choose the correct answer from the options given below

(A) A is false but $\mathbf{R}$ is true

(B) A is true but $\mathbf{R}$ is false

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Open solution

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Q235. JEE Main 2023 (Online) 11th April Morning Shift · MCQ (Single Correct Answer)

Two radioactive elements A and B initially have same number of atoms. The half life of A is same as the average life of B. If λ_A and λ_B are decay constants of A and B respectively, then choose the correct relation from the given options.

Open solution

(A) $\lambda_{\mathrm{A}} = \lambda_{\mathrm{B}} \ln 2$

(B) $\lambda_{\mathrm{A}} \ln 2 = \lambda_{\mathrm{B}}$

(C) $\lambda_{\mathrm{A}} = 2\lambda_{\mathrm{B}}$

(D) $\lambda_{\mathrm{A}} = \lambda_{\mathrm{B}}$

Q236. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

The half life of a radioactive substance is T. The time taken, for disintegrating $\frac{7}{8}$th part of its original mass will be:

Open solution

(A) 8T

(B) 3T

(C) T

(D) 2T

Q237. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

A radio active material is reduced to $1/8$ of its original amount in 3 days. If 8×10^{-3} kg of the material is left after 5 days the initial amount of the material is

Open solution

(A) 64 g

(B) 256 g

(C) 32 g

(D) 40 g

Q238. JEE Main 2023 (Online) 31st January Morning Shift · MCQ (Single Correct Answer)

A free neutron decays into a proton but a free proton does not decay into neutron. This is because

Open solution

(A) neutron is an uncharged particle

(B) neutron has larger rest mass than proton

(C) neutron is a composite particle made of a proton and an electron

(D) proton is a charged particle

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Q239. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A: The nuclear density of nuclides ${}^{10}_{5}\mathrm{B}$, ${}^{6}_{3}\mathrm{Li}$, ${}^{56}_{26}\mathrm{Fe}$, ${}^{20}_{10}\mathrm{Ne}$ and ${}^{209}_{83}\mathrm{Bi}$ can be arranged as $\rho^{\mathrm{N}}_{\mathrm{Bi}} > \rho^{\mathrm{N}}_{\mathrm{Fe}} > \rho^{\mathrm{N}}_{\mathrm{Ne}} > \rho^{\mathrm{N}}_{\mathrm{B}} > \rho^{\mathrm{N}}_{\mathrm{Li}}$

Reason R: The radius R of nucleus is related to its mass number A as $R = R_0 A^{1/3}$, where R_0 is a constant.

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(D) $\mathbf{A}$ is true but $\mathbf{R}$ is false

Q240. JEE Main 2023 (Online) 29th January Evening Shift · MCQ (Single Correct Answer)

Open solution

Substance A has atomic mass number 16 and half life of 1 day. Another substance B has atomic mass number 32 and half life of $\frac{1}{2}$ day. If both A and B simultaneously start undergo radio activity at the same time with initial mass 320 g each, how many total atoms of A and B combined would be left after 2 days.

(A) 1.69×10^{24} (B) 3.38×10^{24} (C) 6.76×10^{23} (D) 6.76×10^{24}

Q241. JEE Main 2023 (Online) 29th January Morning Shift · MCQ (Single Correct Answer)

Open solution

If a radioactive element having half-life of $30\ \mathrm{min}$ is undergoing beta decay, the fraction of radioactive element remains undecayed after $90\ \mathrm{min}$. will be

(A) $\frac{1}{16}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{2}$

Q242. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

Open solution

Consider the following radioactive decay process

$${}^{218}_{84}A \xrightarrow{\alpha} A_1 \xrightarrow{\beta^-} A_2 \xrightarrow{\gamma} A_3 \xrightarrow{\alpha} A_4 \xrightarrow{\beta^+} A_5 \xrightarrow{\gamma} A_6$$

The mass number and the atomic number of A_6 are given by :

(A) 210 and 84 (B) 210 and 80 (C) 211 and 80 (D) 210 and 82

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Q243. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

Read the following statements :

(A) Volume of the nucleus is directly proportional to the mass number.

(B) Volume of the nucleus is independent of mass number.

(C) Density of the nucleus is directly proportional to the mass number.

(D) Density of the nucleus is directly proportional to the cube root of the mass number.

(E) Density of the nucleus is independent of the mass number.

Choose the correct option from the following options.

(A) (A) and (D) only.
(B) (A) and (E) only.
(C) (B) and (E) only.
(D) (A) and (C) only.

Open solution

Q244. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

A radioactive sample decays $\frac{7}{8}$ times its original quantity in 15 minutes. The half-life of the sample is

(A) 5 min
(B) 7.5 min
(C) 15 min
(D) 30 min

Open solution

Q245. JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

The half life period of a radioactive substance is 60 days. The time taken for $\frac{7}{8}$th of its original mass to disintegrate will be :

(A) 120 days
(B) 130 days
(C) 180 days
(D) 20 days

Open solution

Q246. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

The activity of a radioactive material is 6.4×10^{-4} curie. Its half life is 5 days. The activity will become 5×10^{-6} curie after :

(A) 7 days
(B) 15 days
(C) 25 days
(D) 35 days

Open solution

Q247. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

What is the half-life period of a radioactive material if its activity drops to $1/16^{\text{th}}$ of its initial value in 30 years?

(A) 9.5 years
(B) 8.5 years
(C) 7.5 years
(D) 10.5 years

Open solution

Q248. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

Mass numbers of two nuclei are in the ratio of $4 : 3$. Their nuclear densities will be in the ratio of

(A) 4 : 3
(B) $\left(\frac{3}{4}\right)^{\frac{1}{3}}$
(C) 1 : 1
(D) $\left(\frac{4}{3}\right)^{\frac{1}{3}}$

Open solution

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Q249. JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

The disintegration rate of a certain radioactive sample at any instant is 4250 disintegrations per minute. 10 minutes later, the rate becomes 2250 disintegrations per minute. The approximate decay constant is :

(Take $\log_{10} 1.88 = 0.274$)

(A) $0.02\,\text{min}^{-1}$ (B) $2.7\,\text{min}^{-1}$ (C) $0.063\,\text{min}^{-1}$ (D) $6.3\,\text{min}^{-1}$

Open solution

Q250. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

In the following nuclear reaction,

$D \xrightarrow{\alpha} D_1 \xrightarrow{\beta^-} D_2 \xrightarrow{\alpha} D_3 \xrightarrow{\gamma} D_4$

Mass number of D is 182 and atomic number is 74. Mass number and atomic number of D_4 respectively will be _____.

(A) 174 and 71 (B) 174 and 69 (C) 172 and 69 (D) 172 and 71

Open solution

Q251. JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

The activity of a radioactive material is 2.56×10^{-3} Ci. If the half life of the material is 5 days, after how many days the activity will become 2×10^{-5} Ci ?

(A) 30 days (B) 35 days (C) 40 days (D) 25 days

Open solution

Q252. JEE Main 2022 (Online) 28th June Evening Shift · MCQ (Single Correct Answer)

Following statements related to radioactivity are given below :

(A) Radioactivity is a random and spontaneous process and is dependent on physical and chemical conditions.

(B) The number of un-decayed nuclei in the radioactive sample decays exponentially with time.

(C) Slope of the graph of $\log_e$ (no. of undecayed nuclei) Vs. time represents the reciprocal of mean life time (τ).

(D) Product of decay constant (λ) and half-life time ($T_{1/2}$) is not constant.

Choose the most appropriate answer from the options given below :

(A) (A) and (B) only (B) (B) and (D) only (C) (B) and (C) only (D) (C) and (D) only

Open solution

Q253. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

A radioactive nucleus can decay by two different processes. Half-life for the first process is 3.0 hours while it is 4.5 hours for the second process. The effective half-life of the nucleus will be:

(A) 3.75 hours (B) 0.56 hours (C) 0.26 hours (D) 1.80 hours

Open solution

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Q254. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

How many alpha and beta particles are emitted when Uranium $_{92}U^{238}$ decays to lead $_{82}Pb^{206}$?

(A) 3 alpha particles and 5 beta particles

(B) 6 alpha particles and 4 beta particles

(C) 4 alpha particles and 5 beta particles

(D) 8 alpha particles and 6 beta particles

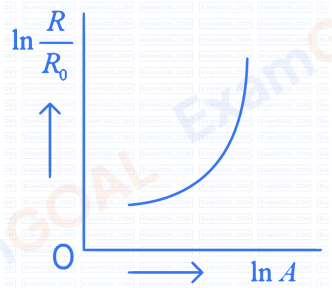
Open solution

Q255. JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

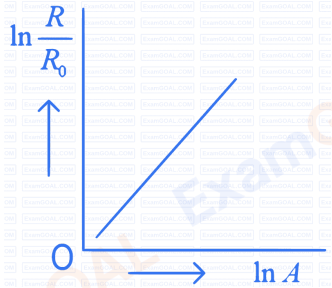
Which of the following figure represents the variation of $\ln\left(\frac{R}{R_0}\right)$ with $\ln A$ (if R = radius of a nucleus and A = its mass number)

Open solution

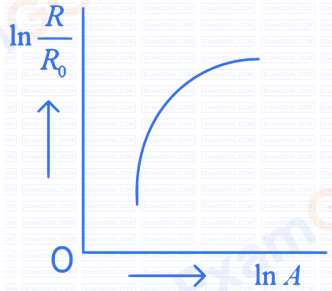
(A)



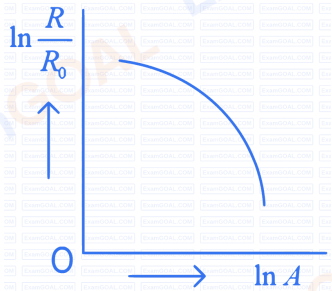
(B)



(C)



(D)



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Q256. JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

The half life period of radioactive element x is same as the mean life time of another radioactive element y. Initially they have the same number of atoms. Then :

(A) x-will decay faster than y.

(B) y-will decay faster than x.

(C) x and y have same decay rate initially and later on different decay rate.

(D) x and y decay at the same rate always.

Open solution

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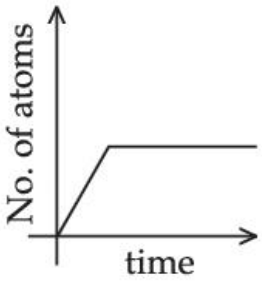
Q257. JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

Open solution

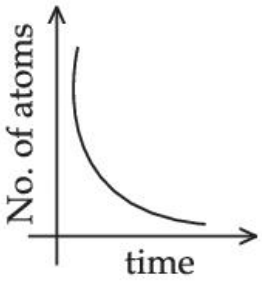
A sample of a radioactive nucleus A disintegrates to another radioactive nucleus B, which in turn disintegrates to some other stable nucleus C. Plot of a graph showing the variation of number of atoms of nucleus B versus time is :

(Assume that at t = 0, there are no B atoms in the sample)

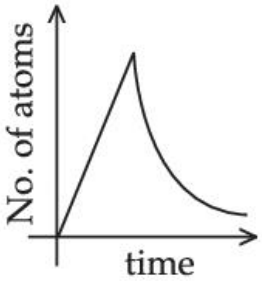
(A)



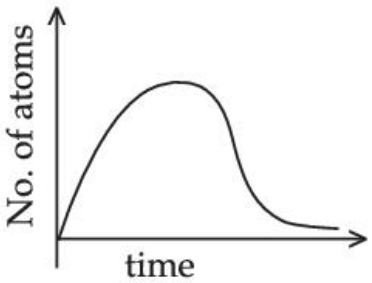
(B)



(C)



(D)



Q258. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

Open solution

There are 10^{10} radioactive nuclei in a given radioactive element, its half-life time is 1 minute. How many nuclei will remain after 30 seconds? $\left(\sqrt{2} = 1.414\right)$

(A) 2×10^{10} (B) 7×10^{9} (C) 10^{5} (D) 4×10^{10}

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Q259. JEE Main 2021 (Online) 26th August Evening Shift · **MCQ (Single Correct Answer)**

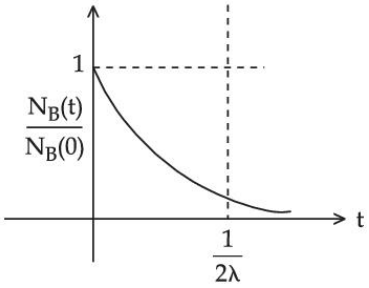
Open solution

At time t = 0, a material is composed of two radioactive atoms A and B, where $N_A(0) = 2N_B(0)$. The decay constant of both kind of radioactive atoms is λ. However, A disintegrates to B and B disintegrates to C. Which of the following figures represents the evolution of $N_B(t) / N_B(0)$ with respect to time t?

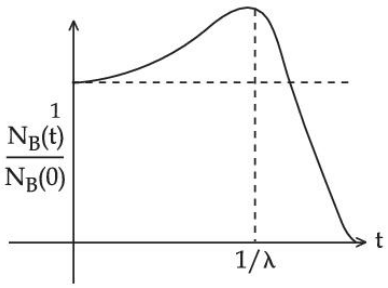
[N_A (0) = No. of A atoms at t = 0

N_B (0) = No. of B atoms at t = 0]

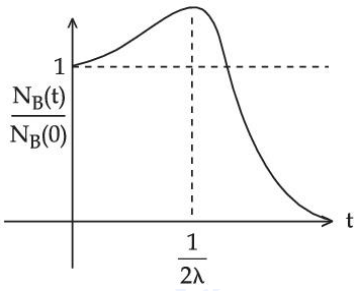
(A)



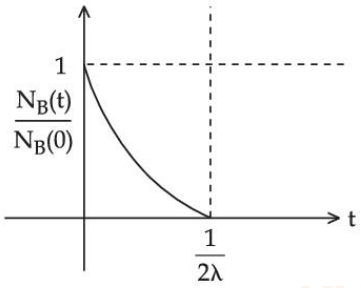
(B)



(C)



(D)



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Q260. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Open solution

Consider the following statements :

A. Atoms of each element emit characteristics spectrum.

B. According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.

C. The density of nuclear matter depends on the size of the nucleus.

D. A free neutron is stable but a free proton decay is possible.

E. Radioactivity is an indication of the instability of nuclei.

Choose the correct answer from the options given below :

(A) A, B, C, D and E (B) A, B and E only (C) B and D only (D) A, C and E only

Q261. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Open solution

If 'f' denotes the ratio of the number of nuclei decayed (N_d) to the number of nuclei at t = 0 (N_0) then for a collection of radioactive nuclei, the rate of change of 'f' with respect to time is given as :

[λ is the radioactive decay constant]

(A) $-\lambda(1-e^{-\lambda t})$ (B) $\lambda(1-e^{-\lambda t})$

(C) $\lambda e^{-\lambda t}$ (D) $-\lambda e^{-\lambda t}$

Q262. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Open solution

Some nuclei of a radioactive material are undergoing radioactive decay. The time gap between the instances when a quarter of the nuclei have decayed and when half of the nuclei have decayed is given as :

(where λ is the decay constant)

(A) $\frac{1}{2}\frac{\ln 2}{\lambda}$ (B) $\frac{\ln 2}{\lambda}$

(C) $\frac{2\ln 2}{\lambda}$ (D) $\frac{\ln\frac{3}{2}}{\lambda}$

Q263. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Open solution

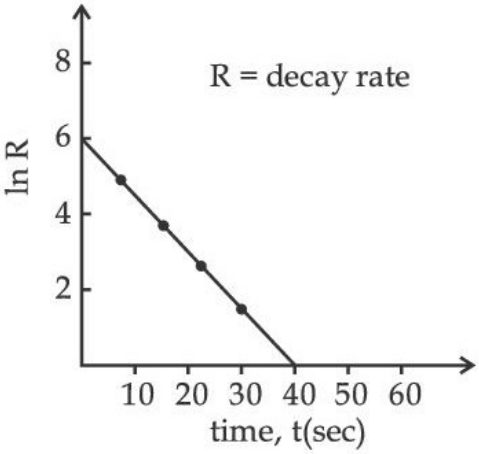
The half-life of ^{198}Au is 3 days. If atomic weight of ^{198}Au is 198 g/mol then the activity of 2 mg of ^{198}Au is [in disintegration/second] :

(A) 2.67×10^{12} (B) 6.06×10^{18} (C) 32.36×10^{12} (D) 16.18×10^{12}

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Q264. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

For a certain radioactive process the graph between ln R and t(sec) is obtained as shown in the figure. Then the value of half life for the unknown radioactive material is approximately :



Open solution

(A) 9.15 sec (B) 6.93 sec (C) 2.62 sec (D) 4.62 sec

Q265. JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

A radioactive material decays by simultaneous emissions of two particles with half lives of 1400 years and 700 years respectively. What will be the time after which one third of the material remains ? (Take ln 3 = 1.1)

Open solution

(A) 740 years (B) 1110 years (C) 700 years (D) 340 years

Q266. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

The decay of a proton to neutron is :

Open solution

(A) always possible as it is associated only with β^+ decay

(B) possible only inside the nucleus

(C) not possible as proton mass is less than the neutron mass

(D) not possible but neutron to proton conversation is possible

Q267. JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

A radioactive sample disintegrates via two independent decay processes having half lives $T_{1/2}^{(1)}$ and $T_{1/2}^{(2)}$ respectively. The effective half-life $T_{1/2}$ of the nuclei is :

Open solution

(A) None of the above

(B) $T_{1/2} = T_{1/2}^{(1)} + T_{1/2}^{(2)}$

(C) $T_{1/2} = \frac{T_{1/2}^{(1)} T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$

(D) $T_{1/2} = \frac{T_{1/2}^{(1)} + T_{1/2}^{(2)}}{T_{1/2}^{(1)} - T_{1/2}^{(2)}}$

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Q268. JEE Main 2021 (Online) 16th March Evening Shift · MCQ (Single Correct Answer)

Calculate the time interval between 33% decay and 67% decay if half-life of a substance is 20 minutes.

(A) 40 minutes (B) 60 minutes (C) 13 minutes (D) 20 minutes

Open solution

Q269. JEE Main 2021 (Online) 16th March Evening Shift · MCQ (Single Correct Answer)

The half-life of Au^{198} is 2.7 days. The activity of 1.50 mg of Au^{198} if its atomic weight is 198 g mol $^{-1}$ is, ($N_a = 6 \times 10^{23}$/mol).

(A) 240 Ci (B) 357 Ci (C) 252 Ci (D) 535 Ci

Open solution

Q270. JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

A radioactive sample is undergoing α decay. At any time t_1, its activity is A and another time t_2, the activity is $\frac{A}{5}$. What is the average life time for the sample?

(A) $\frac{\ln 5}{t_2 - t_1}$ (B) $\frac{\ln(t_2+t_1)}{2}$ (C) $\frac{t_1 - t_2}{\ln 5}$ (D) $\frac{t_2 - t_1}{\ln 5}$

Open solution

Q271. JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

Two radioactive substances X and Y originally have N_1 and N_2 nuclei respectively. Half life of X is half of the half life of Y. After three half lives of Y, number of nuclei of both are equal. The ratio $\frac{N_1}{N_2}$ will be equal to :

(A) $\frac{1}{8}$ (B) $\frac{3}{1}$ (C) $\frac{1}{3}$ (D) $\frac{8}{1}$

Open solution

Q272. JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

A radioactive nucleus decays by two different processes. The half life for the first process is 10 s and that for the second is 100 s. The effective half life of the nucleus is close to :

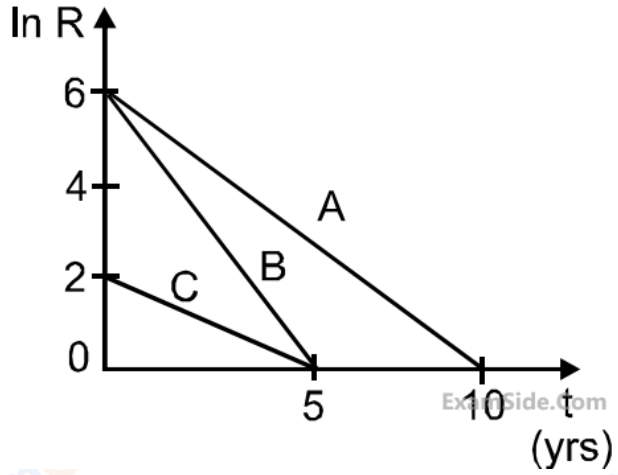
(A) 12 sec (B) 9 sec (C) 55 sec (D) 6 sec

Open solution

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Q273. JEE Main 2020 (Online) 5th September Morning Slot · MCQ (Single Correct Answer)

Activities of three radioactive substances A, B and C are represented by the curves A, B and C, in the figure. Then their half-lives
$T_{\frac{1}{2}}(A) : T_{\frac{1}{2}}(B) : T_{\frac{1}{2}}(C)$ are in the ratio :



Open solution

(A) 3 : 2 : 1 (B) 2 : 1 : 1 (C) 4 : 3 : 1 (D) 2 : 1 : 3

Q274. JEE Main 2020 (Online) 3rd September Morning Slot · MCQ (Single Correct Answer)

In a radioactive material, fraction of active material remaining after time t is 9/16. The fraction that was remaining after t/2 is

(A) $\frac{3}{4}$ (B) $\frac{4}{5}$ (C) $\frac{3}{5}$ (D) $\frac{7}{8}$

Open solution

Q275. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

The activity of a radioactive sample falls from 700 s^{-1} to 500 s^{-1} in 30 minutes. Its half life is close to:

(A) 62 min (B) 66 min (C) 72 min (D) 52 min

Open solution

Q276. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

Half lives of two radioactive nuclei A and B are 10 minutes and 20 minutes, respectively, If initially a sample has equal number of nuclei, then after 60 minutes, the ratio of decayed numbers of nuclei A and B will be :

(A) 9 : 8 (B) 1 : 8 (C) 8 : 1 (D) 3 : 8

Open solution

Q277. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

Two radioactive substances A and B have decay constants 5λ and λ respectively. At t = 0, a sample has the same number of the two nuclei. The time taken for the ratio of the number of nuclei to become $\left(\frac{1}{e}\right)^2$ will be :

(A) $\frac{2}{\lambda}$ (B) $\frac{1}{4\lambda}$ (C) $\frac{1}{2\lambda}$ (D) $\frac{1}{\lambda}$

Open solution

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Q278. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

Two radioactive materials A and B have decay constants 10λ and λ, respectively. It initially they have the same number of nuclei, then the ratio of the number of nuclei of A to that of B will be 1/e after a time :

Open solution

(A) $1/9\lambda$ (B) $11/10\lambda$ (C) $1/10\lambda$ (D) $1/11\lambda$

Q279. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

In a radioactive decay chain, the initial nucleus is $^{232}_{90}$Th. At the end there are 6 α-particles and 4 β-particles which are emitted. If the end nucleus is $^{A}_{Z}$X, A and Z are given by :

Open solution

(A) A = 208; Z = 80 (B) A = 208; Z = 82 (C) A = 200; Z = 81 (D) A = 202; Z = 80

Q280. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

Using a nuclear counter the count rate of emitted particles from a radioactive source is measured. At t = 0 it was 1600 counts per second and t = 8 seconds it was 100 counts per second. The count rate observed, as counts per second, at t = 6 seconds is close to -

Open solution

(A) 200 (B) 150 (C) 400 (D) 360

Q281. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

At a given instant, say t = 0, two radioactive substance A and B have equal activities. the ratio $\frac{R_B}{R_A}$ of their activities after time t itself decays with time t as e^{-3t}. If the half-life of A is $\ln^2$, the half-life of B is :

Open solution

(A) 4ln2 (B) $\frac{\ln 2}{2}$ (C) $\frac{\ln 2}{4}$ (D) 2ln2

Q282. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

A sample of radioactive material A, that has an activity of 10 mCi(1 Ci = 3.7×10^{10} decays/s), has twice the number of nuclei as another sample of a different radioactive materail B which has an activity of 20 mCi. The correct choices for half-lives of A and B would then be respectively :

Open solution

(A) 5 days and 10 days (B) 10 days and 40 days

(C) 20 days and 5 days (D) 20 days and 10 days

Q283. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

At some instant, a radioactive sample S_1 having an activity 5 μCi has twice the number of nuclei as another sample S_2 which has an activity of 10 μCi. The half lives of S_1 and S_2 are :

Open solution

(A) 20 years and 5 years, respectively (B) 20 years and 10years, respectively

(C) 5 years and 20 years, respectively (D) 10 years and 20 years, respectively

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Q284. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

A solution containing active cobalt $^{60}_{27}Co$ having activity of $0.8\ \mu Ci$ and decay constant λ is injected in an animal's body. If $1\ cm^3$ of blood is drawn from the animal's body after 10 hrs of injection, the activity found was 300 decays per minute What is the volume of blood that is flowing in the body ? $\left(\ Ci = 3.7 \times 10^{10}\ \text{decays per second and at } t = 10 \text{ hrs } e^{-\lambda t} = 0.84\right)$

(A) 6 liters (B) 7 liters (C) 4 liters (D) 5 liters

Open solution

Q285. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

A radioactive nucleus A with a half life T, decays into a nucleus B. At t = 0, there is no nucleus B. At sometime t, the ratio of the number of B to that of A is 0.3. Then, t is given by :

(A) $t = \frac{T}{\log(1.3)}$ (B) $t = T\log(1.3)$

(C) $t = \frac{T}{2}\frac{\log 2}{\log 1.3}$ (D) $t = T\frac{\log 1.3}{\log 2}$

Open solution

Q286. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number of nuclei. After 80 minutes, the ratio of decayed number of A and B nuclei will be:

(A) $1:4$ (B) $5:4$ (C) $1:16$ (D) $4:1$

Open solution

Q287. JEE Main 2015 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

Let N_β be the number of β particles emitted by 1 gram of Na^{24} radioactive nuclei (half life = 15 hrs) in 7.5 hours, $N\beta$ is close to (Avogadro number = 6.023×10^{23} / g. mole) :

(A) 6.2×10^{21} (B) 7.5×10^{21} (C) 1.25×10^{22} (D) 1.75×10^{22}

Open solution

Q288. JEE Main 2014 (Online) 19th April Morning Slot · MCQ (Single Correct Answer)

A piece of wood from a recently cut tree shows 20 decays per minute. A wooden piece of same size placed in a museum (obtained from a tree cut many years back) shows 2 decays per minute. If half life of C^{14} is 5730 years, then age of the wooden piece placed in the museum is approximately :

(A) 10439 years (B) 13094 years (C) 19039 years (D) 39049 years

Open solution

Q289. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

A piece of bone of an animal from a ruin is found to have ^{14}C activity of 12 disintegrations per minute per gm of its carbon content. The ^{14}C activity of a living animal is 16 disintegrations per minute per gm. How long ago nearly did the animal die ? (Given half life of ^{14}C is $t_{1/2}$ = 5760 years)

(A) 1672 years (B) 2391 years (C) 3291 years (D) 4453 years

Open solution

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Q290. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

A radioactive nuclei with decay constant 0.5/s is being produced at a constant rate of 100 nuclei/s. If at t = 0 there were no nuclei, the time when there are 50 nuclei is :

(A) 1 s
(B) $2\ln\left(\frac{4}{3}\right)$ s
(C) $\ln 2$ s
(D) $\ln\left(\frac{4}{3}\right)$ s

Open solution

Q291. JEE Main 2013 (Online) 22th April Morning Slot · MCQ (Single Correct Answer)

The half-life of a radioactive element A is the same as the mean-life of another radioactive element B. Initially both substances have the same number of atoms, then :

(A) A and B decay at the same rate always.
(B) A and B decay at the same rate initially.
(C) A will decay at a faster rate than B.
(D) B will decay at a faster rate than A.

Open solution

Q292. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

The counting rate observed from a radioactive source at $t = 0$ was 1600 counts s^{-1}, and $t = 8$ s, it was 100 counts s^{-1}. The counting rate observed as counts s^{-1} at $t = 6$ s will be

(A) 250
(B) 400
(C) 300
(D) 200

Open solution

Q293. AIEEE 2012 (Online) 19th May · MCQ (Single Correct Answer)

The decay constants of a radioactive substance for α and β emission are λ_α and λ_β respectively. If the substance emits α and β simultaneously, then the average half life of the material will be

(A) $\frac{2T_\alpha T_\beta}{T_\alpha + T_\beta}$
(B) $T_\alpha + T_\beta$
(C) $\frac{T_\alpha T_\beta}{T_\alpha + T_\beta}$
(D) $\frac{1}{2}(T_\alpha + T_\beta)$

Open solution

Q294. AIEEE 2012 (Online) 12th May · MCQ (Single Correct Answer)

Which of the following Statements is correct?

(A) The rate of radioactive decay cannot be controlled but that of nuclear fission can be controlled.
(B) Nuclear forces are short range, attractive and charge dependent.
(C) Nuclei of atoms having same number of neutrons are known as isobars.
(D) Wavelength of matter waves is given by de Broglie formula but that of photons is not given by the same formula

Open solution

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Q295. AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

A sample originally contained 10^{20} radioactive atoms, which emit α-particles. The ratio of α particles emitted in the third year to that emitted during the second year is 0.3. How many α particles were emitted in the first year?

(A) 3×10^{18} (B) 3×10^{19} (C) 5×10^{18} (D) 7×10^{19}

Open solution

Q296. AIEEE 2011 · MCQ (Single Correct Answer)

The half life of a radioactive substance is 20 minutes. The approximate time interval $(t_2 - t_1)$ between the time t_2 when $\frac{2}{3}$ of it had decayed and time t_1 when $\frac{1}{3}$ of it had decayed is :

(A) 14 min (B) 20 min (C) 28 min (D) 7 min

Open solution

Q297. AIEEE 2010 · MCQ (Single Correct Answer)

A radioactive nucleus (initial mass number A and atomic number Z emits $3\,\alpha$- particles and 2 positrons. The ratio of number of neutrons to that of protons in the final nucleus will be

(A) $\frac{A-Z-8}{Z-4}$ (B) $\frac{A-Z-4}{Z-8}$ (C) $\frac{A-Z-12}{Z-4}$ (D) $\frac{A-Z-4}{Z-2}$

Open solution

Q298. AIEEE 2007 · MCQ (Single Correct Answer)

In gamma ray emission from a nucleus

(A) only the proton number changes

(B) both the neutron number and the proton number change

(C) there is no change in the proton number and the neutron number

(D) only the neutron number changes

Open solution

Q299. AIEEE 2007 · MCQ (Single Correct Answer)

The half-life period of a radioactive element X is same as the mean life time of another radioactive element Y. Initially they have the same number of atoms. Then

(A) X and Y decay at same rate always (B) X will decay faster than Y

(C) Y will decay faster than X (D) X and Y have same decay rate initially

Open solution

Q300. AIEEE 2006 · MCQ (Single Correct Answer)

The $'rad'$ is the correct unit used to report the measurement of

(A) the ability of a beam of gamma ray photons to produce ions in a target

(B) the energy delivered by radiation to a target

(C) the biological effect of radiation

(D) the rate of decay of radioactive source

Open solution

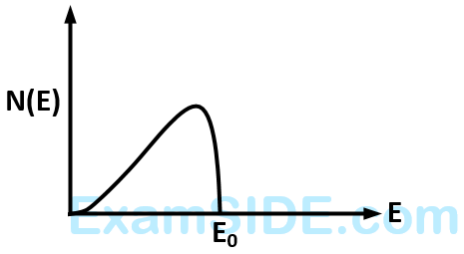
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Q301. AIEEE 2006 · MCQ (Single Correct Answer)

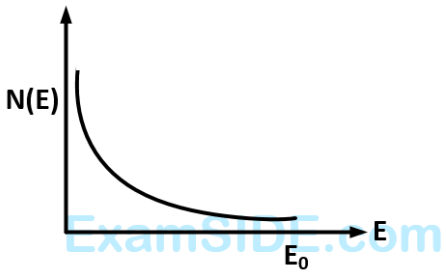
The energy spectrum of β-particles [number $N(E)$ as a function of β-energy E] emitted from a radioactive source is

Open solution

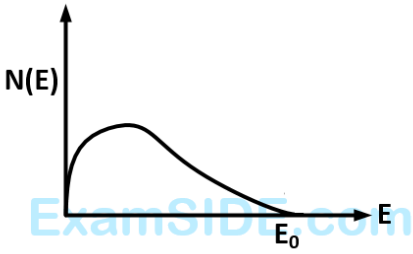
(A)



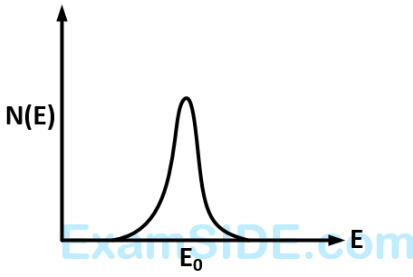
(B)



(C)



(D)



Q302. AIEEE 2005 · MCQ (Single Correct Answer)

If radius of the $^{27}_{13}Al$ nucleus is estimated to be 3.6 fermi then the radius of $^{125}_{52}Te$ nucleus is estimated to be nearly

Open solution

(A) 8 fermi (B) 6 fermi (C) 5 fermi (D) 4 fermi

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Q303. AIEEE 2005 · MCQ (Single Correct Answer)

Starting with a sample of pure ^{66}Cu, $\frac{7}{8}$ of it decays into Zn in 15 minutes. The corresponding half life is

(A) 15 minutes (B) 10 minutes (C) $7\frac{1}{2}$ minutes (D) 5 minutes

Open solution

Q304. AIEEE 2005 · MCQ (Single Correct Answer)

The intensity of gamma radiation from a given source is I. On passing through $36\ mm$ of lead, it is reduced to $\frac{I}{8}$. The thickness of lead which will reduce the intensity to $\frac{I}{2}$ will be

(A) $9mm$ (B) $6mm$ (C) $12mm$ (D) $18mm$

Open solution

Q305. AIEEE 2004 · MCQ (Single Correct Answer)

A nucleus disintegrated into two nuclear parts which have their velocities in the ratio of $2:1$. The ratio of their nuclear sizes will be

(A) $3^{1/2}:1$

(B) $1:2^{1/3}$

(C) $2^{1/3}:1$

(D) $1:3^{1/2}$

Open solution

Q306. AIEEE 2003 · MCQ (Single Correct Answer)

Which of the following radiations has the least wavelength ?

(A) γ - rays (B) β - rays (C) α - rays (D) X - rays

Open solution

Q307. AIEEE 2003 · MCQ (Single Correct Answer)

When a U^{238} nucleus originally at rest, decays by emitting an alpha particle having a speed $'u'$, the recoil speed of the residual nucleus is

(A) $\frac{4\mu}{238}$ (B) $-\frac{4\mu}{234}$ (C) $\frac{4\mu}{234}$ (D) $-\frac{4\mu}{238}$

Open solution

Q308. AIEEE 2003 · MCQ (Single Correct Answer)

A radioactive sample at any instant has its disintegration rate 5000 disintegrations per minute. After 5 minutes, the rate is 1250 disintegrations per minute. Then, the decay constant (per minute) is

(A) $0.4\ ln2$ (B) $0.2\ ln2$ (C) $0.1\ ln2$ (D) $0.8\ ln2$

Open solution

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Q309. AIEEE 2003 · MCQ (Single Correct Answer)

A nucleus with $Z = 92$ emits the following in a sequence:

$$\alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$$

Then Z of the resulting nucleus is

(A) 76 (B) 78 (C) 82 (D) 74

Open solution

Q310. AIEEE 2003 · MCQ (Single Correct Answer)

Which of the following cannot be emitted by radioactive substances during their decay ?

(A) Protons (B) Neutrinoes (C) Helium nuclei (D) Electrons

Open solution

Q311. AIEEE 2002 · MCQ (Single Correct Answer)

At a specific instant emission of radioactive compound is deflected in a magnetic field. The compound can emit

(i) $electrons$ (ii) $protons$

(iii) He^{2+} (iv) $neutrons$

The emission at instant can be

(A) i, ii, iii (B) i, ii, iii, iv (C) iv (D) ii, iii

Open solution

Q312. AIEEE 2002 · MCQ (Single Correct Answer)

If N_0 is the original mass of the substance of half-life period $t_{1/2} = 5$ years, then the amount of substance left after 15 years is

(A) $N_0/8$ (B) $N_0/16$ (C) $N_0/2$ (D) $N_0/4$

Open solution

Q313. JEE Main 2023 (Online) 10th April Morning Shift · Numerical

The decay constant for a radioactive nuclide is 1.5×10^{-5} s^{-1}. Atomic weight of the substance is 60 g $mole^{-1}$, ($N_A = 6 \times 10^{23}$). The activity of 1.0 μg of the substance is _____ $\times 10^{10}$ Bq.

Open solution

Q314. JEE Main 2023 (Online) 30th January Evening Shift · Numerical

A radioactive nucleus decays by two different processes. The half life of the first process is 5 minutes and that of the second process is 30 s. The effective half-life of the nucleus is calculated to be $\frac{\alpha}{11}$ s. The value of α is _____.

Open solution

Q315. JEE Main 2023 (Online) 29th January Morning Shift · Numerical

A radioactive element $^{242}_{92}$X emits two α-particles, one electron and two positrons. The product nucleus is represented by $^{234}_{P}$Y. The value of P is _____.

Open solution

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Q316. JEE Main 2023 (Online) 25th January Evening Shift · **Numerical**

A nucleus disintegrates into two smaller parts, which have their velocities in the ratio 3 : 2. The ratio of their nuclear sizes will be $\left(\frac{x}{3}\right)^{\frac{1}{3}}$. The value of '$x$' is :-

Open solution

Q317. JEE Main 2023 (Online) 24th January Morning Shift · **Numerical**

Assume that protons and neutrons have equal masses. Mass of a nucleon is 1.6×10^{-27} kg and radius of nucleus is $1.5 \times 10^{-15}\ \mathrm{A}^{1/3}$ m. The approximate ratio of the nuclear density and water density is $n \times 10^{13}$. The value of n is _____.

Open solution

Q318. JEE Main 2022 (Online) 29th July Evening Shift · **Numerical**

Two radioactive materials A and B have decay constants 25λ and 16λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of B to that of A will be "e" after a time $\frac{1}{a\lambda}$. The value of a is _____.

Open solution

Q319. JEE Main 2022 (Online) 28th July Morning Shift · **Numerical**

A freshly prepared radioactive source of half life 2 hours 30 minutes emits radiation which is 64 times the permissible safe level. The minimum time, after which it would be possible to work safely with source, will be _____ hours.

Open solution

Q320. JEE Main 2022 (Online) 29th June Evening Shift · **Numerical**

The half life of a radioactive substance is 5 years. After x years a given sample of the radioactive substance gets reduced to 6.25% of its initial value. The value of x is _____.

Open solution

Q321. JEE Main 2022 (Online) 24th June Evening Shift · **Numerical**

A sample contains 10^{-2} kg each of two substances A and B with half lives 4 s and 8 s respectively. The ratio of their atomic weights is 1 : 2. The ratio of the amounts of A and B after 16 s is $\frac{x}{100}$. The value of x is _____.

Open solution

Q322. JEE Main 2021 (Online) 27th July Morning Shift · **Numerical**

A radioactive sample has an average life of 30 ms and is decaying. A capacitor of capacitance 200 μF is first charged and later connected with resistor 'R'. If the ratio of charge on capacitor to the activity of radioactive sample is fixed with respect to time then the value of 'R' should be _____ Ω.

Open solution

Q323. JEE Main 2021 (Online) 25th July Evening Shift · **Numerical**

The nuclear activity of a radioactive element becomes $\left(\frac{1}{8}\right)^{th}$ of its initial value in 30 years. The half-life of radioactive element is _____ years.

Open solution

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Q324. JEE Main 2021 (Online) 20th July Evening Shift · **Numerical**

A radioactive substance decays to $\left(\frac{1}{16}\right)^{th}$ of its initial activity in 80 days. The half life of the radioactive substance expressed in days is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **B**	3. **D**	4. **B**	5. **C**
6. **D**	7. **C**	8. **B**	9. **C**	10. **D**
11. **C**	12. **A**	13. **D**	14. **A**	15. **A**
16. **D**	17. **C**	18. **D**	19. **B**	20. **C**
21. **C**	22. **B**	23. **C**	24. **C**	25. **A**
26. **B**	27. **B**	28. **B**	29. **B**	30. **B**
31. **C**	32. **D**	33. **A**	34. **B**	35. **C**
36. **B**	37. **C**	38. **D**	39. **A**	40. **A**
41. **A**	42. **A**	43. **D**	44. **A**	45. **A**
46. **B**	47. **A**	48. **B**	49. **D**	50. **C**
51. **C**	52. **A**	53. **D**	54. **C**	55. **C**
56. **B**	57. **C**	58. **D**	59. **C**	60. **C**
61. **B**	62. **A**	63. **C**	64. **B**	65. **B**
66. **C**	67. **A**	68. **A**	69. **D**	70. **A**
71. **A**	72. **D**	73. **B**	74. **A**	75. **A**
76. **C**	77. **D**	78. **B**	79. **B**	80. **B**
81. **A**	82. **C**	83. **B**	84. **A**	85. **B**
86. **B**	87. **D**	88. **D**	89. **B**	90. **D**
91. **B**	92. **C**	93. **D**	94. **D**	95. **C**
96. **B**	97. **B**	98. **B**	99. **A**	100. **B**
101. **A**	102. **D**	103. **A**	104. **C**	105. **B**
106. **D**	107. **B**	108. **C**	109. **A**	110. **C**
111. **B**	112. **B**	113. **C**	114. **C**	115. **C**
116. **D**	117. **C**	118. **A**	119. **A**	120. **B**
121. **A**	122. **C**	123. **A**	124. **D**	125. **D**
126. **B**	127. **A**	128. **A**	129. **A**	130. **B**
131. **C**	132. **C**	133. **C**	134. **C**	135. **B**
136. **C**	137. **A**	138. **C**	139. **B**	140. **B**
141. **C**	142. **B**	143. **B**	144. **C**	145. **A**
146. **C**	147. **C**	148. **C**	149. **D**	150. **A**

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151. **C**	152. **B**	153. **C**	154. **B**	155. **C**
156. **D**	157. **A**	158. **D**	159. **B**	160. **C**
161. **B**	162. **C**	163. **D**	164. **D**	165. **B**
166. **C**	167. **D**	168. **B**	169. **C**	170. **B**
171. **A**	172. **B**	173. **A**	174. **A**	175. **D**
176. **B**	177. **C**	178. **A**	179. **C**	180. **C**

Numerical

181. **6**	182. **32**	183. **228**	184. **2**	185. **5**
186. **156**	187. **16**	188. **6588**	189. **727**	190. **208**
191. **7**	192. **32**	193. **27**	194. **4**	195. **1**
196. **6**	197. **121**	198. **17**	199. **144**	200. **236**
201. **5**	202. **2**	203. **4**	204. **2**	205. **3**
206. **3668**	207. **27**	208. **121**	209. **16**	210. **425**
211. **6**	212. **828**	213. **136**	214. **27**	215. **27**
216. **6**	217. **711**	218. **26**	219. **5**	220. **5**
221. **286**	222. **3**	223. **114**	224. **15**	225. **10**
226. **112**	227. **27**	228. **3**	229. **15**	230. **51**
231. **10553**	232. **486**			

Out of syllabus

233. **C**	234. **C**	235. **A**	236. **B**	237. **B**
238. **B**	239. **C**	240. **B**	241. **C**	242. **B**
243. **B**	244. **A**	245. **C**	246. **D**	247. **C**
248. **C**	249. **C**	250. **A**	251. **B**	252. **C**
253. **D**	254. **D**	255. **B**	256. **B**	257. **D**
258. **B**	259. **C**	260. **B**	261. **C**	262. **D**
263. **D**	264. **D**	265. **A**	266. **B**	267. **C**
268. **D**	269. **B**	270. **D**	271. **D**	272. **B**
273. **D**	274. **A**	275. **A**	276. **A**	277. **C**
278. **A**	279. **B**	280. **A**	281. **C**	282. **C**
283. **A**	284. **D**	285. **D**	286. **B**	287. **B**
288. **C**	289. **B**	290. **B**	291. **D**	292. **D**
293. **C**	294. **A**	295. **D**	296. **B**	297. **B**

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298. **C**	299. **C**	300. **C**	301. **C**	302. **B**
303. **D**	304. **C**	305. **B**	306. **A**	307. **C**
308. **A**	309. **B**	310. **A**	311. **A**	312. **A**
313. **15**	314. **300**	315. **87**	316. **2**	317. **11**
318. **9**	319. **15**	320. **20**	321. **25**	322. **150**
323. **10**	324. **20**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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